



KABETE NATIONAL POLYTECHNIC • COMPUTING AND INFORMATICS

How to configure switches and routers in Packet Tracer

A step-by-step Cisco IOS command-line (CLI) workbook for trainees and trainers

Perform Network Design and Management • Unit code 0612 451 07A • ICT Technician Level 6



Set up



VLANs



DHCP



Routing



Security



Troubleshooting



How to use this workbook

How to configure a router interface

Goal: give R1's LAN port its gateway address

```
R1(config)# interface g0/0/0
R1(config-if)# ip address ...
R1(config-if)# no shutdown
```

Select the port
Give it its address
Switch it on

Check it: show ip int brief

Watch out: no shutdown!

3

1

2

4

5

Typing conventions

- **R1(config)#** is the prompt — never type it
- Lines that start with **!** are notes — skip them
- Names, passwords and IPs are lab examples
- Short forms work: **conf t, int g0/0/0, sh ip int br**

1

Title — the skill

Every slide starts with “How to...”.

2

Goal — what and why

What you will achieve, in one line.

3

Dark column — what to type

Type the white text. The green prompt shows the mode you must be in.

4

Light column — what it means

Each command explained in plain English.

5

Check it / Watch out

Prove it worked, and avoid the usual mistake.

Study routine for every slide

Read the goal → build it in Packet Tracer → type the commands → run the check → break it on purpose, then fix it.





How to find your way: the workbook menu

Click a tile to jump to that part. The **home button** at the bottom right of every slide brings you back here.

 01 Packet Tracer basics	 02 The IOS command line	 03 Basic set-up & SSH	 04 VLANs & trunks
 05 Inter-VLAN routing	 06 DHCP	 07 Routing	 08 Security & NAT
 09 STP & EtherChannel	 10 Management & IPv6	 11 Troubleshooting	 12 Practice & quick reference



01

How to get started in Packet Tracer

Build a network on screen: add devices, choose the right cables, open the command line and test with ping.

In this part, learn how to:

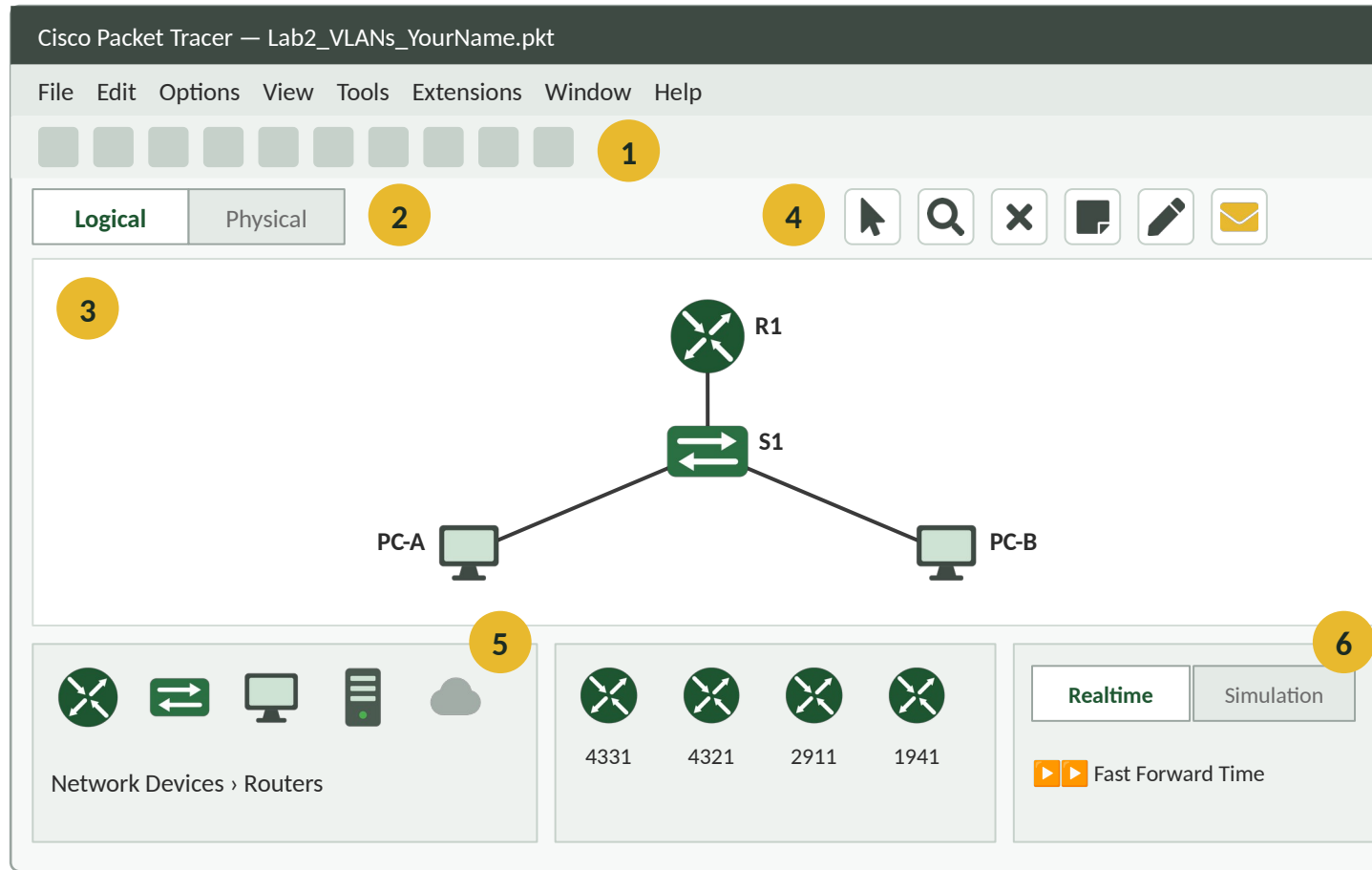
- find your way around the workspace
- add devices and connect the right cable
- open the CLI of a router or switch
- give a PC an address and ping
- watch packets in Simulation mode





How to find your way around the Packet Tracer workspace

Goal: Know where everything is before you build: six areas you will use in every lab.



- 1 **Menu and toolbar**
File › Save As, Undo, Zoom.
- 2 **Logical / Physical tabs**
Stay in Logical: the network diagram.
- 3 **Workspace**
Drag devices here and cable them.
- 4 **Common tools**
Select, Inspect, Delete, Note, Draw, Add Simple PDU.
- 5 **Device-Type Selection box**
Pick a category, then a model.
- 6 **Realtime / Simulation**
Run normally, or step through packets.



Tip: Save often (Ctrl+S): name files like Lab2_VLANs_YourName.pkt





How to add devices and connect them with the right cable

Goal: Build the topology on screen. The right cable makes the link light green; the wrong one leaves it red.

- 1 Add a device**
 Bottom-left: Network Devices › Routers, then drag a **4331** onto the workspace. Switches › **2960** and End Devices › **PC** work the same way.
- 2 Pick a cable**
 Click **Connections** (the lightning-bolt icon), then the cable type you need.
- 3 Connect**
 Click the first device and choose its port, then click the second device and choose its port.
- 4 Read the link lights**
 Green = up. Amber = switch port still starting (wait about 30 s or click Fast Forward Time). Red = down.

Which cable?

-  **Copper Straight-Through** — different device types: PC-switch, switch-router
-  **Copper Cross-Over** — same device types: switch-switch, router-router, PC-PC (and PC-router)
-  **Console** — PC RS 232 port → router or switch Console port, to reach the CLI
-  **Serial DCE / DTE** — router-router WAN link (the router needs a serial module)
-  **Fiber** — fibre ports: long or fast links between switches
-  **Automatic** — Packet Tracer chooses for you — avoid it in assessments

⚠ Watch out: Red light on a router link even with the right cable? Router ports stay off until you type **no shutdown** (part 03).





How to open the command line (CLI) of a router or switch

Goal: Reach the Cisco IOS command line — the way you will configure everything in this workbook.

Quick way — the CLI tab

Click the router or switch → **CLI** tab → click inside the black window and press **Enter**.

Real-life way — a console cable

Console cable: PC **RS 232** → device **Console**. Then PC › Desktop › **Terminal** → OK (keep 9600 bits/s) → press **Enter**.

The Config tab (a shortcut)

Forms that type commands for you, shown at the bottom as **Equivalent IOS Commands**. Learn from it, but practise in the CLI.

```
● ● ● R1 › CLI (end of the start-up messages)

  --- System Configuration Dialog ---

Would you like to enter the initial
configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#
```

Answer **no** — you will configure by hand. **Router>** = user mode (look only); **enable** takes you to **Router#**.



Watch out: Nothing on screen? Click inside the CLI window and press Enter — the device may still be starting up.





How to give a PC an IP address and test it with ping

Goal: Every PC needs an address, mask and default gateway before it can talk. Then prove it with ping.

PC-A › Desktop › IP Configuration

○ DHCP ● **Static**

IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.50.10

PC-A › Command Prompt (output shortened)

```
C:\>ipconfig
IPv4 Address.....: 192.168.1.10
Subnet Mask.....: 255.255.255.0
Default Gateway...: 192.168.1.1

C:\>ping 192.168.1.1
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
...
Packets: Sent = 4, Received = 4, Lost = 0

C:\>tracert 192.168.20.10
! lists every router on the way
```

- 1 Click the PC → **Desktop** → **IP Configuration**.
- 2 Choose **Static**, type address, mask and gateway (or **DHCP** — part 06).
- 3 Close it, open **Command Prompt** and test.



Watch out: The first ping may lose one reply while ARP finds the MAC address. That is normal — ping again.



Tip: No typing: **Add Simple PDU** (closed envelope) → click source → click destination → read the PDU list.

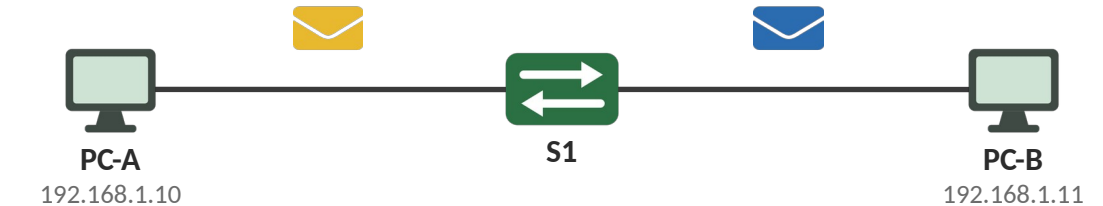




How to watch packets travel in Simulation mode

Goal: Slow the network down and see each packet hop by hop — the best way to understand ARP, DHCP and routing.

- 1 Click **Simulation** (bottom-right, next to Realtime).
- 2 **Edit Filters** → tick only what you want to see, e.g. **ICMP** and **ARP**.
- 3 Make traffic: **Add Simple PDU** from PC-A to PC-B, or ping from the Command Prompt.
- 4 Press **Capture/Forward** for one step, or **Play** to run.
- 5 Click an envelope (or an Event List row) → the **OSI Model** tab explains each layer.
- 6 Click **Realtime** to go back to normal.



Time (s)	Last device	At device	Type
0.000	--	PC-A	ARP
0.001	PC-A	S1	ARP
0.002	S1	PC-B	ARP
0.003	PC-B	S1	ARP
0.005	PC-A	S1	ICMP



Check it: You should see ARP first (“who has 192.168.1.11?”), then the ICMP echo request and reply.



Watch out: Nothing moves? Tick the protocol in **Edit Filters**, then press **Capture/Forward**.



02

How to work in the Cisco IOS command line

Every Cisco router and switch speaks IOS. Learn its modes, its built-in help and how to read its messages before you configure anything.

In this part, learn how to:

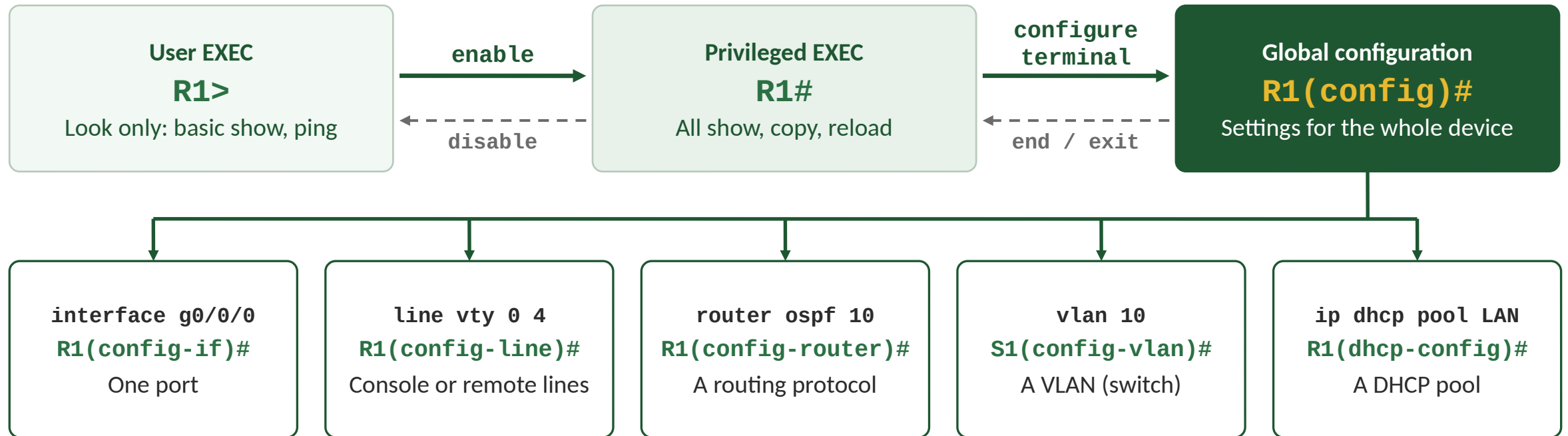
- move between the command modes
- get help and type less
- read IOS error messages
- name interfaces correctly





How to move between the IOS command modes

Goal: IOS is like a building: each room (mode) allows different commands. The prompt tells you which room you are in.



Typed in global configuration mode, these commands open the smaller “rooms”. **exit** = one level back; **end** or Ctrl+Z = straight back to **R1#**.



Check it: Type **?** at any prompt: IOS lists exactly what that mode accepts.



Watch out: Read the prompt before you type. Most errors are the right command in the wrong mode.





How to get help and type less in the CLI

Goal: IOS has built-in help and shortcuts. Use them and you will type less and make fewer mistakes.

Key or command	What it does	Example
?	List every command or option you can type at this point	R1# show ?
co?	List the commands that start with those letters (no space before ?)	configure connect copy
Tab	Complete a partly typed word	conf + Tab gives configure
Short forms	Any abbreviation that is unique works	conf t, int g0/0/0, sh ip int br
↑ ↓ arrows	Recall the previous or next command	Edit it, then press Enter
end or Ctrl+Z	Jump straight back to privileged mode (#)	R1(config-if)# end → R1#
Ctrl+Shift+6	Stop a running ping, traceroute or name lookup	When a typo freezes the screen
do	Run a show or copy command without leaving config mode	do show ip int brief
no	Undo (reverse) a command	no shutdown, no ip address



Tip: Learn with **?**: type **show ?** or **interface ?** and IOS lists exactly what it expects next.





How to read IOS error messages

Goal: Error messages are clues, not failures. Each one tells you what kind of mistake you made.

You typed	IOS replied	Why — and the fix
R1> configure terminal	% Invalid input detected at '^' marker.	You are in user mode (>). Type enable first.
R1(config)# show running-config	% Invalid input detected at '^' marker.	show is not a config command. Use do show running-config .
R1(config-if)# ip address 192.168.1.1	% Incomplete command.	Something is missing — here the mask. Type ? to see what comes next.
R1# c	% Ambiguous command: "c"	Several commands start with c. Type more letters.
R1# shwo	Translating "shwo"...domain server (255.255.255.255)	IOS thinks the typo is a computer name. Press Ctrl+Shift+6; prevent it with no ip domain-lookup .
R1(config-if)# ip address 192.168.1.5 255.255.255.0	% 192.168.1.0 overlaps with GigabitEthernet0/0/0	Two router ports can't share a subnet. Give this port a different one.

**Tip:** The ^ marker points at the first character IOS did not understand — look there first.





How to name the interfaces you type

Goal: Every command that touches a port needs its exact name. The name tells you the type and where the port sits.

GigabitEthernet

/

0

/

0

/

0

type (short: g, fa, s)

slot

module bay

port

Not sure of the name?
Run **show ip interface brief** — it lists every port your device really has.

Device (Packet Tracer model)	Interface names	Type it as
Router 4331	GigabitEthernet0/0/0 and 0/0/1. 0/0/2 is an empty slot: power off, add a GLC-T module, power on	g0/0/0
Router 1941 / 2901 / 2911	GigabitEthernet0/0 and 0/1 (the 2911 also has 0/2)	g0/0
Router + serial module	4331 + NIM-2T: Serial0/1/0 – 0/1/1. 1941 + HWIC-2T: Serial0/0/0 – 0/0/1	s0/1/0
Switch 2960-24TT, multilayer 3560-24PS	FastEthernet0/1 – 0/24 and GigabitEthernet0/1 – 0/2	fa0/1, g0/1
Virtual interfaces	Vlan1, Vlan99 (SVI) • Port-channel1 • g0/0/1.10 (sub-interface)	int vlan 99



03

How to set up a new router or switch

The “starter pack” for every Cisco device: name it, secure it, address it, save it — and manage it safely with SSH. (Lab 1)

In this part, learn how to:

- do the basic set-up on any device
- save, view and erase a configuration
- address a router interface
- read show ip interface brief
- give a switch a management IP
- configure many ports at once
- set up SSH remote access





How to do the basic set-up on any router or switch (1 of 2)

Goal: Name the device, protect privileged mode and add a legal warning. Do this first on every new device.

● ● ● R1 — type the white text, then press Enter	What it does
Router> enable	Move from user mode (>) to privileged mode (#). Nothing can be configured from >.
Router# configure terminal	Enter global configuration mode. From now on, each command changes the live configuration at once.
Router(config)# hostname R1	Name the device. The prompt becomes R1(config)# , so you always know which device you are on.
R1(config)# no ip domain-lookup	Stop IOS treating a typo as a computer name to look up (it freezes the screen for about 30 seconds).
R1(config)# enable secret Kabete@2026	The password for enable , stored as a secure hash. Never use the old enable password — it is stored as plain text.
R1(config)# banner motd # Authorised access only! #	A legal warning shown to everyone who connects. The # signs mark where the message starts and ends.



Check it: Type **end**, then **disable**, then **enable** again: R1 must now ask for the password.



Watch out: If your message contains #, use another start/end sign, e.g. **banner motd \$... \$**.





How to do the basic set-up on any router or switch (2 of 2)

Goal: Lock the console port, keep messages tidy, hide passwords and save. The same commands work on a switch.

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# line console 0	Open the settings of the console port (the light-blue cable). The prompt becomes (config-line)# .
R1(config-line)# password Cons@1234	Set the console password.
R1(config-line)# login	Make the console ask for that password. Without login , the password is never requested.
R1(config-line)# logging synchronous	Stop system messages from interrupting your typing.
R1(config-line)# exec-timeout 5 0	Log out a session that is idle for 5 minutes 0 seconds.
R1(config-line)# exit R1(config)# service password-encryption	Back to global mode, then hide the console and VTY passwords in show running-config (weak: it only stops onlookers).
R1(config)# end R1# copy running-config startup-config	Back to privileged mode, then save your work.



Check it: Type **exit** to log out, then press Enter: the banner appears and the console asks for a password.



Watch out: Passwords are case-sensitive. Record them in your lab notes — a forgotten secret means password recovery.





How to save, view and erase a configuration

Goal: The live configuration (RAM) is lost at power-off. The saved one (NVRAM) is loaded at every start-up.

● ● ● R1 — type the white text, then press Enter	What it does
R1# <code>copy running-config startup-config</code>	Save the live configuration to NVRAM so it survives a restart. Press Enter to accept [startup-config] .
R1# <code>write memory</code>	A shorter way to make exactly the same save (wr works too).
R1# <code>show running-config</code>	Show the live configuration. Space = next page, Enter = next line, Q = quit.
R1# <code>show startup-config</code>	Show the saved configuration. Anything missing here is not saved yet.
R1# <code>erase startup-config</code> S1# <code>delete flash:vlan.dat</code>	Wipe the saved configuration to start a lab again. A switch keeps its VLANs in vlan.dat — delete that too.
R1# <code>reload</code>	Restart the device. After an erase, answer no to “Save?” so it starts empty.



Check it: After saving, **show startup-config** shows your hostname and passwords.



Watch out: Anything not saved is lost when the device reloads or loses power — save after every step that works.





How to configure a router interface with an IP address

Goal: Give R1’s LAN port an address so the PCs on that LAN can use it as their default gateway (192.168.1.0/24).

● ● ● R1 — type the white text, then press Enter	What it does
<pre>R1(config)# interface g0/0/0</pre>	Select the port (full name GigabitEthernet0/0/0). The prompt becomes (config-if)# .
<pre>R1(config-if)# description LAN to S1 - ICT Lab</pre>	A label for people. It changes nothing, but helps whoever troubleshoots later.
<pre>R1(config-if)# ip address 192.168.1.1 255.255.255.0</pre>	The port’s address and subnet mask. PCs on this LAN use 192.168.1.1 as their default gateway.
<pre>R1(config-if)# no shutdown</pre>	Switch the port on. Router ports start “administratively down” — the number-one cause of red links.
<pre>R1(config-if)# exit</pre>	Back to global configuration mode.

Check it: `show ip interface brief` → G0/0/0 192.168.1.1 up up. From PC-A, `ping 192.168.1.1` gets replies.

Watch out: Each router port needs its own subnet, and the PCs’ gateway must match this address exactly.





How to read show ip interface brief

Goal: The most-used command in this workbook. Each row is a port; two columns tell you what is wrong.

R1 > CLI (shortened)

```
R1# show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/0/1  unassigned     YES unset  administratively down down
```

Status	Protocol	What it means	Usual fix
up	up	Working: cable and port are fine (Layers 1 and 2)	—
administratively down	down	The port is switched off (never enabled, or shutdown typed)	no shutdown
down	down	No signal: cable missing or wrong type, or the far end is off	Check the cable and the other end
up	down	Signal OK, but the two ends disagree (e.g. encapsulation)	Compare the settings at both ends

Tip: Method: **manua**l**** = you typed the address, **DHCP** = it was leased, **unset** = no address yet.





How to give a switch a management IP address

Goal: A switch forwards frames without any IP. Give it one only so you can manage it over the network (ping, SSH).

● ● ● S1 — type the white text, then press Enter	What it does
<pre>S1(config)# interface vlan 1</pre>	Open the switch virtual interface (SVI) of VLAN 1 — a virtual port inside the switch.
<pre>S1(config-if)# ip address 192.168.1.2 255.255.255.0</pre>	The switch's management address, in the same subnet as the PCs of that VLAN.
<pre>S1(config-if)# no shutdown</pre>	SVIs start switched off. It turns up/up once a port in that VLAN is up.
<pre>S1(config-if)# exit</pre>	Back to global configuration mode.
<pre>S1(config)# ip default-gateway 192.168.1.1</pre>	Where the switch sends replies for other subnets: R1's LAN address.



Check it: `show ip interface brief` → Vlan1 192.168.1.2 up up. PC-A: `ping 192.168.1.2` gets replies.



Watch out: A Layer 2 switch has no routing table: without `ip default-gateway` it can only reply inside its own subnet.





How to configure many switch ports at once

Goal: `interface range` applies the same commands to a group of ports — faster, and fewer mistakes.

● ● ● S1 — type the white text, then press Enter	What it does
<code>S1(config)# interface range fa0/1 - 24</code>	Select ports Fa0/1 to Fa0/24 together. The prompt becomes (config-if-range)# .
<code>S1(config-if-range)# description USER PORT</code>	Every selected port gets the same label.
<code>S1(config-if-range)# interface range fa0/1 - 5, fa0/10</code>	Mix ranges and single ports, separated by commas.
<code>S1(config-if-range)# interface range fa0/3 - 24, g0/2</code> <code>S1(config-if-range)# shutdown</code>	Switch off every port you are not using, so nobody can plug in. (Lab 1 uses Fa0/1, Fa0/2 and G0/1.)
<code>S1(config-if-range)# interface fa0/3</code> <code>S1(config-if)# no shutdown</code>	Re-enable a single port when a new user needs it.



Check it: `show ip interface brief` lists the unused ports as administratively down.



Watch out: Don't shut the uplink to the router (G0/1 in Lab 1), or you lose contact with the rest of the network.





How to set up SSH for secure remote access (1 of 2)

Goal: SSH lets you manage a device over the network, encrypted. It needs a hostname, a domain name, a user and RSA keys.

● ● ● R1 — type the white text, then press Enter	What it does
<code>R1(config)# ip domain-name knp.ac.ke</code>	Needed to name the encryption keys (R1.knp.ac.ke). RSA keys cannot be made without it.
<code>R1(config)# username netadmin privilege 15 secret N3t@dmin26</code>	A personal admin account stored on the device. Privilege 15 = full control straight after login.
<code>R1(config)# username helpdesk privilege 1 secret H3lp@desk26</code>	A help-desk account: it logs in at R1> (look only) and needs the enable secret to go further.
<code>R1(config)# crypto key generate rsa general-keys modulus 1024</code>	Create the RSA keys that encrypt SSH; this also turns SSH on. Use at least 1024 bits (2048 on real kit).
<code>R1(config)# ip ssh version 2</code>	Allow only SSH version 2 — version 1 has known weaknesses.



Check it: `show ip ssh` → “SSH Enabled - version 2.0”.



Watch out: “Please define a hostname other than Router” or “... a domain-name first”? Set them, then generate the keys again.





How to set up SSH for secure remote access (2 of 2)

Goal: Tell the remote-access (VTY) lines to accept SSH only, using the local accounts. Then test from a PC.

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# line vty 0 4	Open the 5 virtual lines used by remote sessions (a 2960 switch has 16: line vty 0 15).
R1(config-line)# transport input ssh	Accept SSH only. Telnet is refused because it sends passwords as plain text.
R1(config-line)# login local	Ask for a username and password from the local accounts (the username commands).
R1(config-line)# exec-timeout 5 0	Log out remote sessions that are idle for 5 minutes.
R1(config-line)# end R1# copy running-config startup-config	Back to privileged mode and save.
C:\> ssh -l netadmin 192.168.1.1 <i>! then type the password (it is not shown)</i>	Test from PC-A › Command Prompt. netadmin lands at R1# ; helpdesk lands at R1> .



Check it: From PC-A, SSH works and **telnet 192.168.1.1** fails. On a switch, secure **line vty 0 15** the same way.



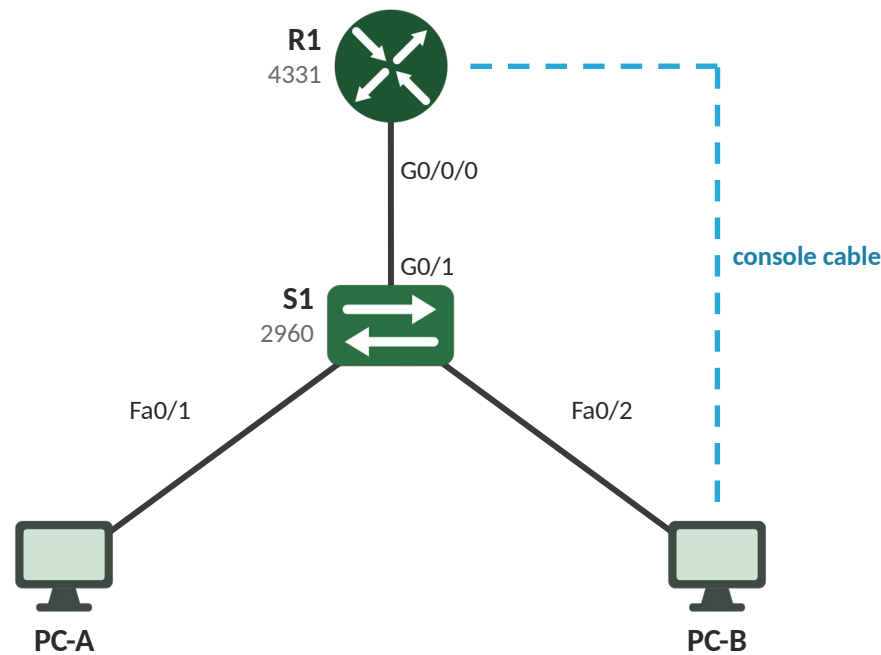
Watch out: Use **login local**, not **login:** with **login** IOS wants a line password you never set, so every login fails.





How to practise: Lab 1 — basic router and switch set-up

Goal: Build this network and apply everything in this part. Then prove it works with the checks at the bottom.



Device	Interface	IP address / mask	Gateway
R1	G0/0/0	192.168.1.1 /24	—
S1	VLAN 1	192.168.1.2 /24	192.168.1.1
PC-A	NIC	192.168.1.10 /24	192.168.1.1
PC-B	NIC	192.168.1.11 /24	192.168.1.1

Tasks

1. Build and cable it (straight-through; console cable PC-B → R1).
2. Basic set-up on R1 and S1: hostname, secret, console, banner.
3. Address R1 G0/0/0, S1 VLAN 1 and both PCs; shut unused ports.
4. SSH on R1 and S1 (user netadmin); save both devices.
5. Prove it: all pings work, SSH works, Telnet is refused.

Use the console cable for the first set-up of R1 — the way a brand-new real router is configured.

04

How to separate a network with VLANs

Split one switched network into departments — STAFF, STUDENTS, FINANCE — and carry them all between switches over trunks. (Lab 2)

In this part, learn how to:

- understand VLANs before you configure
- plan the VLAN lab
- create and name VLANs
- put ports into VLANs
- configure trunk links
- manage a switch through VLAN 99
- check VLANs and trunks
- change VLANs safely





How to understand VLANs before you configure them

Goal: A VLAN turns one physical switch into several separate “rooms”, each with its own subnet.

Without VLANs, a switch is one big room: a broadcast from any PC reaches every PC.

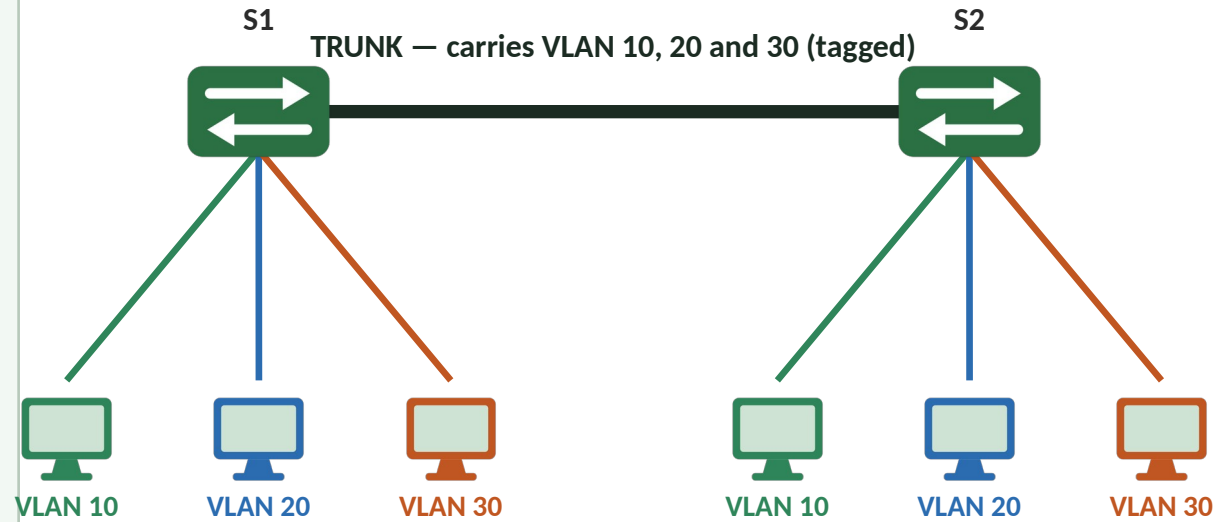
VLANs make separate rooms. A port in VLAN 10 only talks to other VLAN 10 ports.

One VLAN = one subnet. STAFF = 192.168.10.0/24, STUDENTS = 192.168.20.0/24.

Access port — carries one VLAN; for a PC, printer or access point.

Trunk port — carries many VLANs between switches (and to a router); each frame gets an 802.1Q tag with its VLAN number.

Between VLANs you need a router or a Layer 3 switch (part 05).



A frame on the trunk:

Dest MAC	Src MAC	802.1Q tag: VLAN 10	Data
----------	---------	---------------------	------

The receiving switch reads the tag, removes it and delivers the frame only to VLAN 10 ports.

Think of it like this

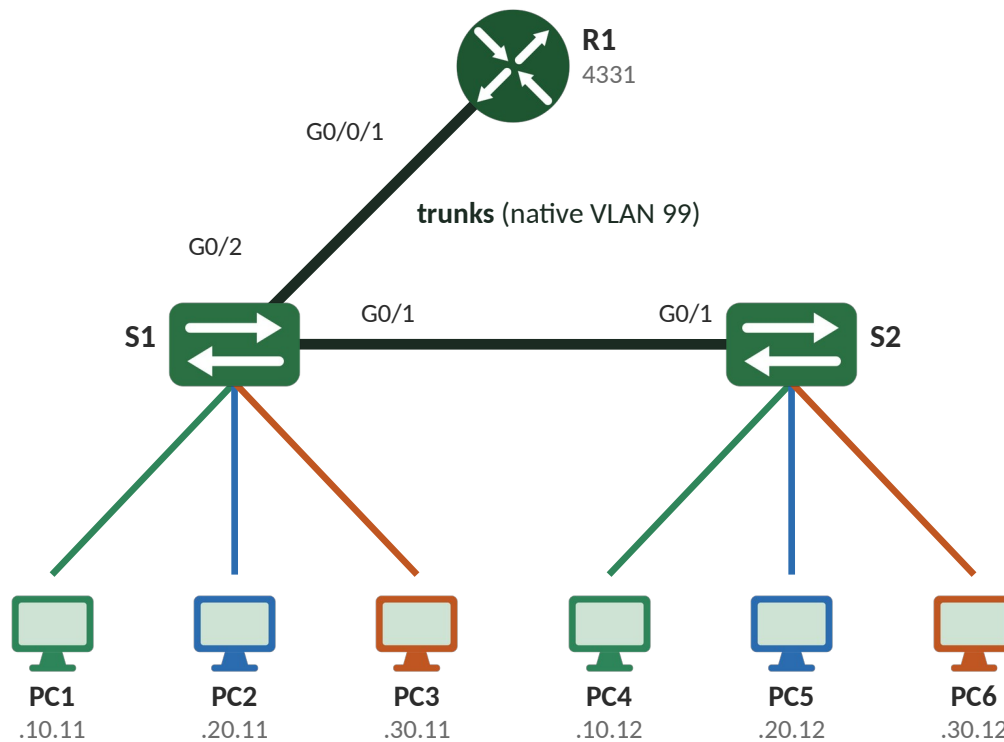
One building (switch), separate rooms (VLANs), one shared lift (trunk) that keeps everyone's badge on (the tag).





How to plan the VLAN lab (Lab 2 topology)

Goal: Plan before you type: every VLAN gets a number, a name, a set of ports and its own subnet.



PC labels show the end of the address: .10.11 = 192.168.10.11 /24

VLAN	Name	Access ports	Subnet	Gateway
10	STAFF	Fa0/1 – 10	192.168.10.0 /24	192.168.10.1
20	STUDENTS	Fa0/11 – 20	192.168.20.0 /24	192.168.20.1
30	FINANCE	Fa0/21 – 24	192.168.30.0 /24	192.168.30.1
99	MGMT (native)	none — trunks only	192.168.99.0 /24	192.168.99.1

Good to know

- Switch management addresses: S1 = 192.168.99.2, S2 = 192.168.99.3.
- Gateways are R1 sub-interfaces (part 05) — the .1 of each VLAN.
- The same VLANs must exist on S1 and S2.
- Cables: switch-switch cross-over; switch-router straight-through.





How to create and name VLANs

Goal: Create the four VLANs of the plan on S1 — then do exactly the same on S2.

● ● ● S1 — type the white text, then press Enter	What it does
<pre>S1(config)# vlan 10 S1(config-vlan)# name STAFF</pre>	Create VLAN 10, enter VLAN mode, and give it a name people understand. Names show in show vlan brief .
<pre>S1(config-vlan)# vlan 20 S1(config-vlan)# name STUDENTS</pre>	Create the next VLAN straight from VLAN mode — no need to exit first.
<pre>S1(config-vlan)# vlan 30 S1(config-vlan)# name FINANCE</pre>	VLAN 30 for the finance office.
<pre>S1(config-vlan)# vlan 99 S1(config-vlan)# name MGMT</pre>	VLAN 99 for managing the switches; it will also be the trunks' native VLAN.
<pre>S1(config-vlan)# end</pre>	Back to privileged mode. The VLANs are applied when you leave VLAN mode.



Check it: **show vlan brief** lists 10 STAFF, 20 STUDENTS, 30 FINANCE and 99 MGMT (no ports yet).



Watch out: VLAN 1 is the default: it cannot be renamed or deleted. Repeat these commands on S2!





How to put switch ports into a VLAN (access ports)

Goal: Place each user port in its department's VLAN. A PC plugged into that port joins that VLAN's subnet.

● ● ● S1 — type the white text, then press Enter	What it does
<code>S1(config)# interface range fa0/1 - 10</code>	Select the ten STAFF ports.
<code>S1(config-if-range)# switchport mode access</code>	Fix them as access ports: one VLAN, for end devices. They can no longer turn themselves into trunks.
<code>S1(config-if-range)# switchport access vlan 10</code>	Put the ports in VLAN 10 (STAFF).
<code>S1(config-if-range)# interface range fa0/11 - 20</code> <code>S1(config-if-range)# switchport mode access</code> <code>S1(config-if-range)# switchport access vlan 20</code>	The same three steps for the STUDENTS ports.
<code>S1(config-if-range)# interface range fa0/21 - 24</code> <code>S1(config-if-range)# switchport mode access</code> <code>S1(config-if-range)# switchport access vlan 30</code>	And for the FINANCE ports. Then do the same on S2.



Check it: show vlan brief → Fa0/1-10 beside VLAN 10, Fa0/11-20 beside 20, Fa0/21-24 beside 30.



Watch out: A missing VLAN is created for you ("% Access VLAN does not exist. Creating vlan 10") — but with no name.





How to configure a trunk link

Goal: Links between switches, and to the router, must carry every VLAN. Make them 802.1Q trunks — at both ends.

● ● ● S1 — type the white text, then press Enter	What it does
<code>S1(config)# interface range g0/1 - 2</code>	Select both uplinks: G0/1 goes to S2, G0/2 goes to R1.
<code>S1(config-if-range)# switchport mode trunk</code>	Make them trunks: they carry all VLANs, tagging each frame with its VLAN number (802.1Q).
<code>S1(config-if-range)# switchport trunk native vlan 99</code>	Untagged frames now belong to VLAN 99 instead of VLAN 1 — safer. Both ends of a link must agree.
<code>S1(config-if-range)# switchport trunk allowed vlan 10,20,30,99</code>	Only these VLANs may cross the trunk.
<code>S2(config)# interface g0/1</code> <code>S2(config-if)# switchport mode trunk</code> <code>S2(config-if)# switchport trunk native vlan 99</code> <code>S2(config-if)# switchport trunk allowed vlan 10,20,30,99</code>	The same on S2's uplink to S1 — a trunk has two ends!



Check it: `show interfaces trunk` → G0/1 and G0/2: mode on, status trunking, native vlan 99.



Watch out: On a 3560, type `switchport trunk encapsulation dot1q` before `switchport mode trunk`.





How to manage a switch through VLAN 99

Goal: Move each switch's management address out of VLAN 1 into the MGMT VLAN, away from user traffic.

● ● ● S1 — type the white text, then press Enter	What it does
<pre>S1(config)# interface vlan 99</pre>	Open the SVI of the management VLAN.
<pre>S1(config-if)# ip address 192.168.99.2 255.255.255.0</pre>	S1's management address (S2 uses 192.168.99.3).
<pre>S1(config-if)# no shutdown</pre>	Switch the SVI on.
<pre>S1(config-if)# exit S1(config)# ip default-gateway 192.168.99.1</pre>	R1's VLAN 99 sub-interface (part 05) is the way out for management traffic.
<pre>S1(config)# interface vlan 1 S1(config-if)# shutdown</pre>	Optional: switch off the unused VLAN 1 interface so it is not used by mistake.



Check it: `show ip interface brief` → Vlan99 192.168.99.2 up up (once a trunk carrying VLAN 99 is up).



Watch out: Vlan99 stays down until VLAN 99 exists on the switch and a port or trunk carrying it is up.





How to check VLANs and trunks

Goal: Two show commands prove the VLAN design is built correctly. Learn to read them line by line.

S1 > CLI (columns shortened)

S1# show vlan brief

VLAN	Name	Status	Ports
1	default	active	
10	STAFF	active	Fa0/1, Fa0/2, Fa0/3 ...
20	STUDENTS	active	Fa0/11, Fa0/12 ...
30	FINANCE	active	Fa0/21, Fa0/22, Fa0/23, Fa0/24
99	MGMT	active	

S1# show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	99
Gig0/2	on	802.1q	trunking	99

Port Vlan allowed on trunk

Gig0/1	10,20,30,99
Gig0/2	10,20,30,99

Ports beside the right VLAN

The access ports are correct. Trunk ports never appear in this list — that is normal.

trunking + native vlan 99

The trunk is up and both ends agree on the native VLAN.

Vlans allowed on trunk

Only these VLANs can cross. A missing VLAN here means its PCs can't reach the other switch.

Same-VLAN ping fails? Check the VLAN exists on both switches.





How to change, add or remove VLANs safely

Goal: Networks change. Edit VLAN membership and trunk lists without cutting off users who are already working.

● ● ● S1 — type the white text, then press Enter	What it does
<code>S1(config-if)# switchport trunk allowed vlan add 40</code>	Add VLAN 40 to a trunk's list. Without the word add , the list is replaced and the other VLANs stop crossing!
<code>S1(config-if)# switchport trunk allowed vlan remove 30</code>	Take just VLAN 30 off the trunk.
<code>S1(config)# interface fa0/5</code> <code>S1(config-if)# switchport access vlan 20</code>	Move one port to another VLAN: re-type the command with the new number.
<code>S1(config-if)# no switchport access vlan</code>	Send the port back to VLAN 1 (the default).
<code>S1(config)# no vlan 30</code>	Delete VLAN 30. Its ports become inactive — move them to another VLAN first.



Check it: Run **show vlan brief** and **show interfaces trunk** after every change.



Watch out: Ports left in a deleted VLAN go silent: their PCs lose the network with no error message.



05

How to let VLANs talk to each other

VLANs are separate subnets, so traffic between them must be routed. Use router-on-a-stick (Lab 2) or a Layer 3 switch.

In this part, learn how to:

- choose an inter-VLAN routing method
- configure router-on-a-stick
- follow a packet between VLANs
- route on a Layer 3 switch





How to choose an inter-VLAN routing method

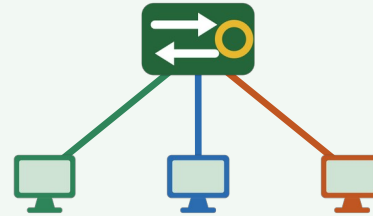
Goal: Three ways to route between VLANs. Labs use the first two; know all three for the assessment.



Router-on-a-stick

One router port + one trunk; a sub-interface per VLAN (**g0/0/1.10**).

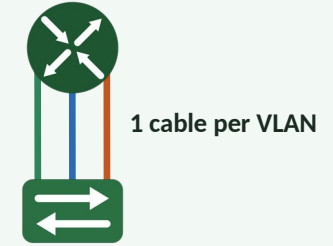
- ✓ Cheap — one cable
- ✗ All inter-VLAN traffic shares one link
- Use:** small sites and labs (Lab 2)



Layer 3 switch (SVIs)

The switch routes inside itself; **interface vlan 10** is VLAN 10's gateway.

- ✓ Fast and scalable
- ✗ Needs a multilayer switch (3560)
- Use:** campus core / distribution



One router port per VLAN

Each VLAN gets its own physical router port and its own cable.

- ✓ Simple to understand
- ✗ Runs out of router ports quickly
- Use:** rarely — old designs

Tip: Whatever the method, each PC's default gateway is the router or SVI address in its own VLAN (the .1).





How to configure router-on-a-stick (1 of 2)

Goal: R1 routes between the VLANs over one trunk to S1. Each VLAN gets a sub-interface on G0/0/1.

R1 — type the white text, then press Enter	What it does
<pre>R1(config)# interface g0/0/1 R1(config-if)# no shutdown</pre>	Switch on the physical port that connects to S1's trunk. It gets no IP address itself.
<pre>R1(config-if)# interface g0/0/1.10</pre>	Create sub-interface 10, a virtual port inside G0/0/1. Matching the number to the VLAN avoids confusion.
<pre>R1(config-subif)# description STAFF gateway</pre>	A label for people.
<pre>R1(config-subif)# encapsulation dot1Q 10</pre>	The key command: this sub-interface handles frames tagged VLAN 10. Type it before the IP address.
<pre>R1(config-subif)# ip address 192.168.10.1 255.255.255.0</pre>	The default gateway of every STAFF PC.

Check it: `show ip interface brief` → G0/0/1 and G0/0/1.10 are both up up.

Watch out: IOS refuses the IP address until **encapsulation dot1Q** is set — always type it first.





How to configure router-on-a-stick (2 of 2)

Goal: Add a sub-interface for every other VLAN — including the native management VLAN — then test across VLANs.

● ● ● R1 — type the white text, then press Enter	What it does
<pre>R1(config-subif)# interface g0/0/1.20 R1(config-subif)# encapsulation dot1Q 20 R1(config-subif)# ip address 192.168.20.1 255.255.255.0</pre>	The gateway of the STUDENTS VLAN.
<pre>R1(config-subif)# interface g0/0/1.30 R1(config-subif)# encapsulation dot1Q 30 R1(config-subif)# ip address 192.168.30.1 255.255.255.0</pre>	The gateway of the FINANCE VLAN.
<pre>R1(config-subif)# interface g0/0/1.99 R1(config-subif)# encapsulation dot1Q 99 native R1(config-subif)# ip address 192.168.99.1 255.255.255.0</pre>	The MGMT gateway. native = untagged frames belong here; it must match the trunk's native VLAN on S1.
<pre>R1(config-subif)# end R1# copy running-config startup-config</pre>	Finish and save.



Check it: show ip route connected → four C routes (VLANs 10, 20, 30, 99). PC1 (VLAN 10) pings PC2 (VLAN 20).



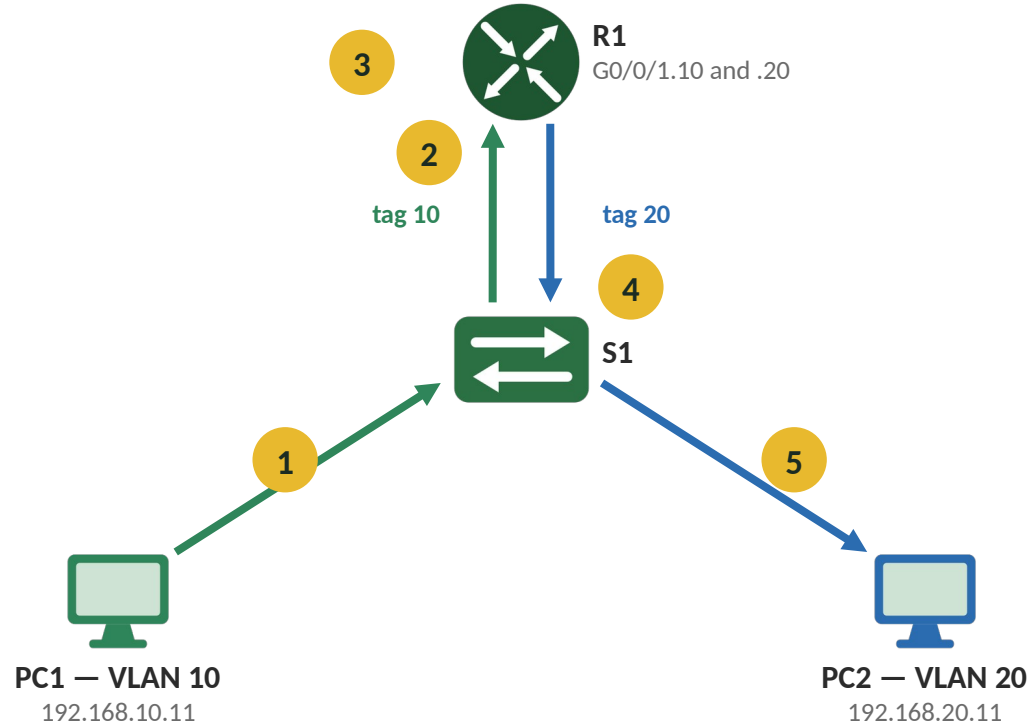
Watch out: S1's port to R1 (G0/2) must be a trunk, and each PC's gateway must be the .1 of its own VLAN.





How to follow a packet from VLAN 10 to VLAN 20

Goal: Understand what router-on-a-stick really does: the packet goes up the trunk and comes back down it.



Tip: See it yourself: Simulation mode, filter ICMP, ping PC1 → PC2, and open the envelope on the trunk to see the 802.1Q tag.





How to route between VLANs on a Layer 3 switch

Goal: A multilayer switch (3560) can be the gateway for every VLAN itself — faster than router-on-a-stick.

● ● ● MLS1 — type the white text, then press Enter	What it does
MLS1(config)# ip routing	Turn on routing inside the switch. Without it the SVIs cannot route between VLANs.
MLS1(config)# vlan 10 MLS1(config-vlan)# name STAFF MLS1(config-vlan)# vlan 20 MLS1(config-vlan)# name STUDENTS MLS1(config-vlan)# exit	Create the VLANs on the multilayer switch too.
MLS1(config)# interface vlan 10 MLS1(config-if)# ip address 192.168.10.1 255.255.255.0 MLS1(config-if)# no shutdown	The SVI of VLAN 10 is now the default gateway of the STAFF PCs.
MLS1(config-if)# interface vlan 20 MLS1(config-if)# ip address 192.168.20.1 255.255.255.0 MLS1(config-if)# no shutdown	The same for VLAN 20 (STUDENTS). Access ports are set exactly as in part 04.



Check it: **show ip route** lists both VLAN subnets as C (connected via Vlan10 and Vlan20); the PCs ping each other.



Watch out: On a 3560, **no switchport** turns a port into a routed port, and trunks need **switchport trunk encapsulation dot1q** first.



06

How to hand out IP addresses automatically with DHCP

Let the router lease addresses, masks, gateways and DNS servers to every PC — for one LAN, for every VLAN, and across subnets with a relay. (Lab 4, built on the Lab 2 file)

In this part, learn how to:

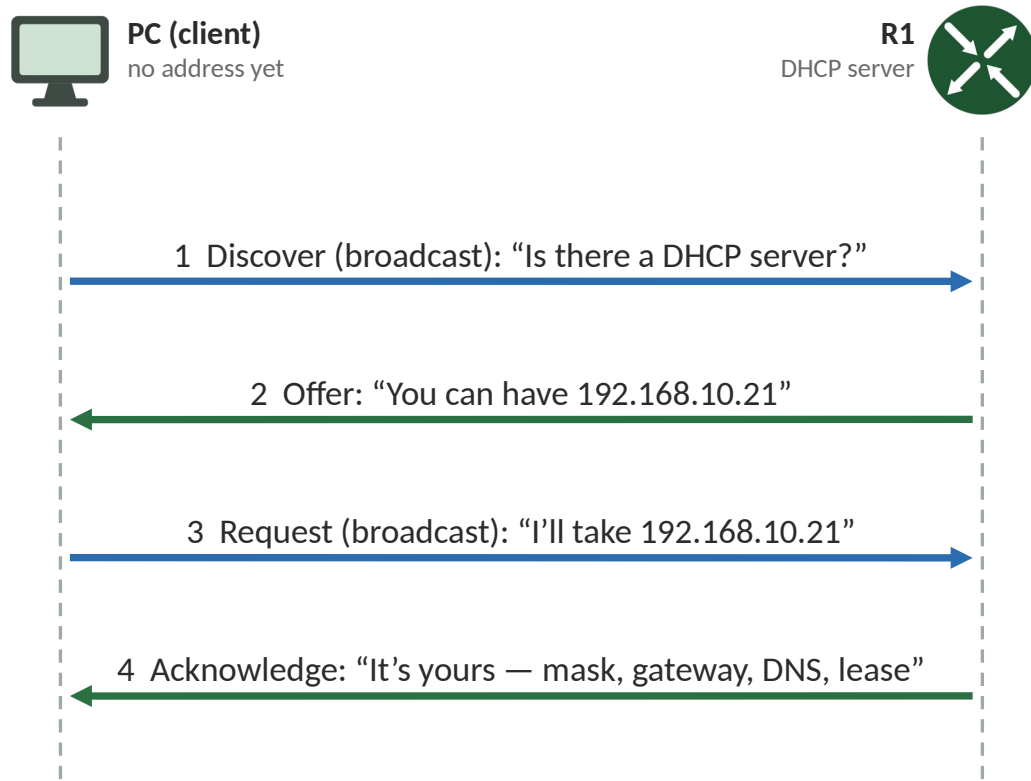
- understand the DORA conversation
- make a router a DHCP server
- give every VLAN its own pool
- check the leases
- relay DHCP to a server
- fix common DHCP problems





How to understand DHCP: the DORA conversation

Goal: A PC with no address asks for one. Four messages later it has an address, mask, gateway and DNS server.



What the PC receives

IP address, subnet mask, default gateway and DNS server (plus domain name and lease time).

DHCP or static?

DHCP: PCs, laptops, phones. **Static:** routers, switches, servers, printers — keep their addresses out of the pool.

A good plan for each /24 subnet

.1 = gateway • .2 - .20 = static devices • .21 - .254 = DHCP pool



Watch out: A PC showing 169.254.x.x got no DHCP answer — it gave itself an automatic (APIPA) address.





How to configure a router as a DHCP server (1 of 2)

Goal: R1 will lease addresses to the STAFF PCs in VLAN 10 (192.168.10.0/24), keeping .1-.20 for static devices.

R1 — type the white text, then press Enter	What it does
<pre>R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20</pre>	Never lease .1 to .20 — they are kept for the gateway, servers, printers and switches. Do this before the pool.
<pre>R1(config)# ip dhcp pool STAFF-POOL</pre>	Create a pool (you choose the name) and enter (dhcp-config)# mode.
<pre>R1(dhcp-config)# network 192.168.10.0 255.255.255.0</pre>	The subnet to lease from: .21 to .254 after the exclusions.
<pre>R1(dhcp-config)# default-router 192.168.10.1</pre>	The default gateway given to clients: the router’s own address on that LAN.
<pre>R1(dhcp-config)# dns-server 192.168.50.10</pre>	The DNS server clients use to turn names into IP addresses.

Check it: show **ip dhcp pool** → STAFF-POOL, with the number of leased addresses.

Watch out: A wrong **default-router** lets PCs get an address but never leave their own subnet.





How to configure a router as a DHCP server (2 of 2)

Goal: Finish the pool, then switch the PCs to DHCP and check that they lease their addresses.

R1 — type the white text, then press Enter	What it does
<pre>R1(dhcp-config)# domain-name knp.local</pre>	The DNS suffix added to short names (www becomes www.knp.local).
<pre>R1(dhcp-config)# exit</pre>	Back to global configuration mode.
<pre>! On each PC: Desktop > IP Configuration > DHCP</pre>	The PC asks for an address. “DHCP request successful” appears and the fields fill in by themselves.
<pre>C:\> ipconfig /all</pre>	Shows the leased address, mask, gateway, DNS server and the PC’s MAC (Physical Address).
<pre>R1# show ip dhcp binding</pre>	One line per lease: the IP given out and the MAC of the PC that has it.



Check it: Each STAFF PC gets 192.168.10.21 or higher, gateway 192.168.10.1 and DNS 192.168.50.10.



Watch out: Select **DHCP** in IP Configuration first; **ipconfig /release** and **/renew** then get a fresh lease.





How to give every VLAN its own DHCP pool

Goal: One router can serve every VLAN. It picks the pool whose network matches the sub-interface the request came in on.

● ● ● R1 — type the white text, then press Enter	What it does
<pre>R1(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.20 R1(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.20</pre>	Keep the first 20 addresses of each VLAN for static devices.
<pre>R1(config)# ip dhcp pool STUDENT-POOL R1(dhcp-config)# network 192.168.20.0 255.255.255.0 R1(dhcp-config)# default-router 192.168.20.1 R1(dhcp-config)# dns-server 192.168.50.10</pre>	The pool for VLAN 20. Its gateway is sub-interface G0/0/1.20.
<pre>R1(dhcp-config)# ip dhcp pool FINANCE-POOL R1(dhcp-config)# network 192.168.30.0 255.255.255.0 R1(dhcp-config)# default-router 192.168.30.1 R1(dhcp-config)# dns-server 192.168.50.10</pre>	The pool for VLAN 30. Its gateway is G0/0/1.30.



Check it: PCs in VLANs 10, 20 and 30 each lease an address from their own subnet; **show ip dhcp binding** lists them all.



Watch out: No pool for VLAN 99: switches and routers keep static addresses.





How to check DHCP leases on the router and the PC

Goal: Match what the router handed out with what the PC received. The MAC address links the two.

R1 > CLI (columns shortened)

R1# show ip dhcp binding

IP address	Hardware address	Type
192.168.10.21	0060.4721.6A1B	Automatic
192.168.20.21	00D0.97C4.3A02	Automatic
192.168.30.21	0001.C7A3.11F5	Automatic

R1# show ip dhcp pool

Pool STAFF-POOL :

Total addresses	: 254
Leased addresses	: 1
Excluded addresses	: 20

PC1 > Command Prompt (shortened)

C:\>ipconfig /all

FastEthernet0 Connection:(default port)
 Physical Address.....: 0060.4721.6A1B
 IPv4 Address.....: 192.168.10.21
 Subnet Mask.....: 255.255.255.0
 Default Gateway.....: 192.168.10.1
 DHCP Servers.....: 192.168.10.1
 DNS Servers.....: 192.168.50.10

Match the MAC

The Hardware address in the binding is the PC's Physical Address: now you know who has which IP.

Leased addresses

Counts up as PCs join. Still 0 while PCs wait?
 Their requests are not reaching R1.

Renew a lease

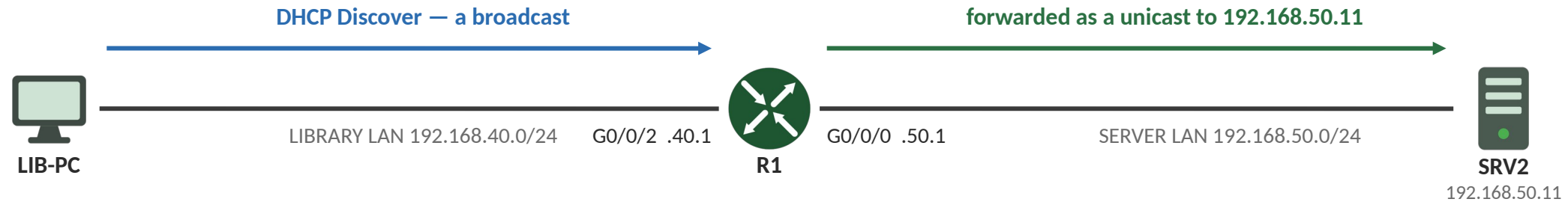
On the PC: **ipconfig /release**, then
ipconfig /renew (DHCP must be selected).





How to relay DHCP from another subnet (ip helper-address)

Goal: The LIBRARY LAN (192.168.40.0/24) gets its addresses from SRV2, a DHCP server in another subnet.



● ● ● R1 — type the white text, then press Enter

```
R1(config)# interface g0/0/2
R1(config-if)# ip address 192.168.40.1 255.255.255.0
R1(config-if)# no shutdown
```

```
R1(config-if)# ip helper-address 192.168.50.11
```

What it does

The LIBRARY port — the clients' gateway. (On a 4331, add a GLC-T module to G0/0/2 first.)

Routers drop broadcasts. This catches the DHCP broadcast and forwards it to SRV2, stamped with 192.168.40.1 so SRV2 picks the right pool.

On SRV2: Services › DHCP

Service **On** • Pool Name **LIB-POOL** • Default Gateway **192.168.40.1** • DNS Server **192.168.50.10** • Start IP **192.168.40.100** • Mask **255.255.255.0** • Maximum Users **50** → **Add**

Watch out

Helper goes on the port facing the **clients**. SRV2 needs gateway 192.168.50.1 to reply.





How to fix common DHCP problems

Goal: Most DHCP faults are not in the pool at all — they are VLAN, trunk, gateway or relay mistakes.

What you see	Most likely cause	How to fix it
PC address is 169.254.x.x	No DHCP reply reached the PC	Check the pool network , the port's VLAN, the trunk's allowed VLANs, and the helper address
PC has an address but cannot leave its subnet	Wrong default-router in the pool	Set it to the router's address in that subnet, then renew the lease
PC gets an address from the wrong subnet	Its switch port is in the wrong VLAN	switchport access vlan with the right VLAN number
Two devices with the same address	A static device's address is not excluded	Add ip dhcp excluded-address for it, then renew the PCs
Websites work by IP but not by name	Missing or wrong dns-server	Fix dns-server in the pool and renew the lease
Relay clients get nothing	Helper on the wrong port, or the server has no route back	Helper on the client-side port; give the server a default gateway



Tip: Prove every fix: **ipconfig /renew** on the PC, then **show ip dhcp binding** on the router.



07

How to connect networks with routing

Teach routers where every network is — by hand with static routes, or automatically with OSPF, RIP and EIGRP. (Lab 3)

In this part, learn how to:

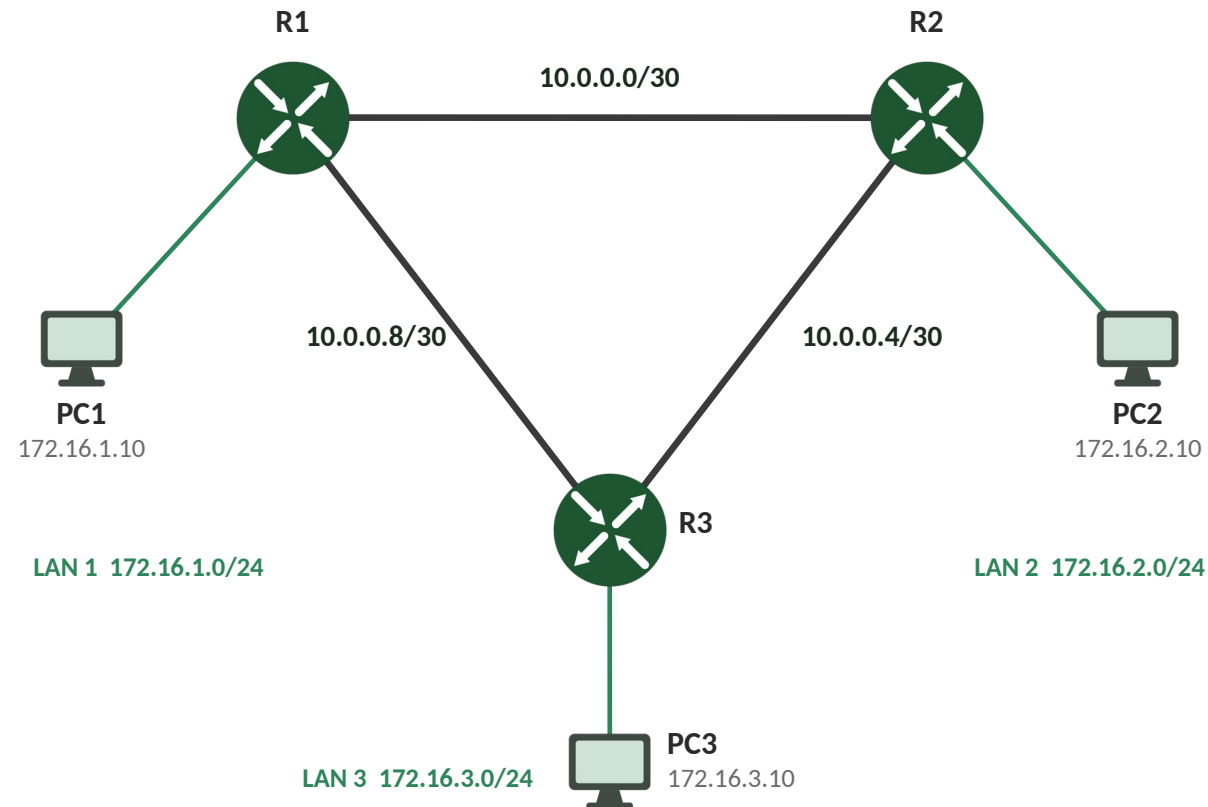
- set up the routing lab
- read a routing table
- configure static and default routes
- configure single-area OSPF
- configure RIPv2 and EIGRP
- check and troubleshoot routing





How to set up the routing lab (Lab 3 topology)

Goal: Three routers, three LANs, three point-to-point links. Every link is its own /30 subnet.



Router	G0/0/0 (LAN)	G0/0/1	G0/0/2
R1	172.16.1.1 /24	10.0.0.1 /30 → R2	10.0.0.9 /30 → R3
R2	172.16.2.1 /24	10.0.0.2 /30 → R1	10.0.0.5 /30 → R3
R3	172.16.3.1 /24	10.0.0.6 /30 → R2	10.0.0.10 /30 → R1

Order of work

1. Router-router links: cross-over cables. Add a GLC-T module to each G0/0/2.
2. Address every interface and **no shutdown**; PCs use their router's .1 as gateway.
3. Ping each neighbour across every /30 link before any routing.
4. Then: static → OSPF → RIP → EIGRP. Remove each before the next, e.g. **no router ospf 10**.





How to read a routing table

Goal: `show ip route` is the router's map. Each line tells you how a network was learned and which way to send packets.

● ● ● R1 › CLI (shortened — OSPF running)

R1# `show ip route`

Codes: L - local, C - connected, S - static, R - RIP,
O - OSPF, D - EIGRP, * - candidate default

Gateway of last resort is **not set**

```
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
C    172.16.1.0/24 is directly connected, GigabitEthernet0/0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0/0
O    172.16.2.0/24 [110/2] via 10.0.0.2, 00:01:12, GigabitEthernet0/0/1
O    172.16.3.0/24 [110/2] via 10.0.0.10, 00:01:12, GigabitEthernet0/0/2
```

Code letter

How the route was learned: C connected, L the router's own address, S static, O OSPF, R RIP, D EIGRP, S* default.

[110/2]

[administrative distance / metric]. Lower distance = more trusted source; lower metric = better path.

via 10.0.0.2 • G0/0/1

The next-hop router, and the port the packet leaves by.

Gateway of last resort

The default route. "not set" = packets to unknown networks are dropped.

Administrative distance — the lower number wins:

0 Connected

1 Static

90 EIGRP

110 OSPF

120 RIP

If two methods know the same network, only the lowest distance goes into the table — that is why each lab part removes the previous method.





How to configure static routes

Goal: Tell each router by hand where the networks it is not connected to are (Lab 3, part A).

● ● ● R1 / R2 / R3 — type the white text, then press Enter	What it does
R1(config)# ip route 172.16.2.0 255.255.255.0 10.0.0.2	To reach LAN 2, send packets to R2 (10.0.0.2, its address on our shared link). Parts: network, mask, next hop.
R1(config)# ip route 172.16.3.0 255.255.255.0 10.0.0.10	To reach LAN 3, send packets to R3.
R1(config)# ip route 172.16.2.0 255.255.255.0 10.0.0.10 5	A floating static route: a backup path to LAN 2 via R3 with distance 5, used only if the main route (distance 1) fails.
R2(config)# ip route 172.16.1.0 255.255.255.0 10.0.0.1 R2(config)# ip route 172.16.3.0 255.255.255.0 10.0.0.6 R2(config)# ip route 172.16.1.0 255.255.255.0 10.0.0.6 5	R2 needs routes too — every router needs a route there and back — plus its own floating backup to LAN 1 via R3.
R3(config)# ip route 172.16.1.0 255.255.255.0 10.0.0.9 R3(config)# ip route 172.16.2.0 255.255.255.0 10.0.0.5	R3's routes to LAN 1 and LAN 2.



Check it: show ip route static shows S routes; PC1 pings PC2 and PC3. Shut R1 G0/0/1: the floating routes take over.



Watch out: The next hop is the neighbour's address on the shared link — never your own address or the neighbour's LAN address.





How to configure a default route

Goal: A default route catches every destination the router doesn't know — usually “all Internet traffic: go to the ISP”.

● ● ● R1 / R2 — type the white text, then press Enter	What it does
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1	“Anything I don’t know, send to the ISP router 203.0.113.1.” 0.0.0.0 0.0.0.0 matches every address.
R2(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1	A branch router with only one way out needs just this one route instead of many static routes.
R1(config)# router ospf 10 R1(config-router)# default-information originate	With OSPF running, R1 shares its default route, so the other routers learn it automatically (as O*E2).
R1# show ip route	Look for S* 0.0.0.0/0 [1/0] via 203.0.113.1 and “Gateway of last resort is 203.0.113.1”.



Check it: From a PC, **tracert** to an Internet server: the trace passes R1 and then the ISP router.



Watch out: Point the default route only at the real exit. Two routers defaulting to each other bounce packets until they expire.





How to configure OSPF (single area) (1 of 2)

Goal: OSPF lets routers find every network by themselves and reroute when a link fails (Lab 3, part B — R1 shown).

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# router ospf 10	Start OSPF process 10. The number is local — neighbours don't need to match it.
R1(config-router)# router-id 1.1.1.1	A unique ID for R1 (R2 = 2.2.2.2, R3 = 3.3.3.3) that makes outputs easy to read.
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0	Run OSPF on the port in 172.16.1.0/24 and put it in area 0. 0.0.0.255 is the wildcard of 255.255.255.0.
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0 R1(config-router)# network 10.0.0.8 0.0.0.3 area 0	Run OSPF on both /30 links (wildcard 0.0.0.3).
R1(config-router)# passive-interface g0/0/0	No OSPF hellos to the PCs, but the LAN is still advertised.



Check it: **show ip ospf neighbor** → 2.2.2.2 and 3.3.3.3 in FULL state; **show ip route ospf** → O routes.



Watch out: Remove the static routes first (**no ip route ...**) — with distance 1 they would hide the OSPF routes.





How to configure OSPF (single area) (2 of 2)

Goal: The same pattern on R2 and R3: their own router ID, their own LAN and their two links.

R2 > CLI

```
R2(config)# router ospf 10
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 172.16.2.0 0.0.0.255 area 0
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 10.0.0.4 0.0.0.3 area 0
R2(config-router)# passive-interface g0/0/0
```

R3 > CLI

```
R3(config)# router ospf 10
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 172.16.3.0 0.0.0.255 area 0
R3(config-router)# network 10.0.0.4 0.0.0.3 area 0
R3(config-router)# network 10.0.0.8 0.0.0.3 area 0
R3(config-router)# passive-interface g0/0/0
```

Work out the wildcard: 255.255.255.255 – subnet mask

Prefix	Subnet mask	Wildcard
/24	255.255.255.0	0.0.0.255
/26	255.255.255.192	0.0.0.63
/30	255.255.255.252	0.0.0.3

Check it on every router

- **show ip protocols** lists the networks OSPF is running on
- **show ip ospf neighbor** shows two neighbours in FULL state
- Each PC can ping the other two PCs





How to check OSPF neighbours and routes

Goal: Two commands prove OSPF works: the neighbours are FULL and the remote LANs appear as O routes.

R1 › CLI

R1# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:35	10.0.0.2	GigabitEthernet0/0/1
3.3.3.3	1	FULL/BDR	00:00:33	10.0.0.10	GigabitEthernet0/0/2

R1# show ip route ospf

```

0    172.16.2.0/24 [110/2] via 10.0.0.2, 00:02:10, GigabitEthernet0/0/1
0    172.16.3.0/24 [110/2] via 10.0.0.10, 00:02:10, GigabitEthernet0/0/2
0    10.0.0.4/30 [110/2] via 10.0.0.2, 00:02:10, GigabitEthernet0/0/1
                        [110/2] via 10.0.0.10, 00:02:10, GigabitEthernet0/0/2
  
```

FULL

The routers have swapped their complete maps. INIT, 2WAY or no line at all? Check areas, wildcards and passive ports. DR/BDR is just the neighbour's role on that link.

[110/2]

OSPF's distance (110) and cost (2): 1 for the link to R2 plus 1 for R2's LAN — each Gigabit link costs 1.

Two "via" lines

Two paths with the same cost: OSPF shares traffic across both (equal-cost load balancing).





How to configure RIP version 2

Goal: RIP is the simplest routing protocol — fine for small networks. First remove OSPF: **no router ospf 10** (part C).

● ● ● R1 — type the white text, then press Enter	What it does
<code>R1(config)# router rip</code>	Start RIP and enter (config-router)# mode.
<code>R1(config-router)# version 2</code>	Use RIPv2: it sends subnet masks (needed for /30 and /24 subnets) and uses multicast.
<code>R1(config-router)# no auto-summary</code>	Advertise the real subnets, not the summary 172.16.0.0/16.
<code>R1(config-router)# network 172.16.0.0</code>	Run RIP on the ports in 172.16.x.x. RIP takes classful network numbers — no mask.
<code>R1(config-router)# network 10.0.0.0</code>	Run RIP on both 10.0.0.x links.
<code>R1(config-router)# passive-interface g0/0/0</code>	No RIP updates to the PCs; the LAN is still advertised.



Check it: `show ip route rip` → R 172.16.2.0/24 [120/1] via 10.0.0.2 (120 = RIP's distance, 1 = one hop).



Watch out: **network** means “run RIP on my ports in this network”, not “route to it”. RIP gives up after 15 hops.





How to configure EIGRP

Goal: EIGRP is Cisco's fast-converging protocol. First remove RIP: **no router rip** (part D).

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# router eigrp 100	Start EIGRP in autonomous system 100. Every neighbour must use the same number.
R1(config-router)# eigrp router-id 1.1.1.1	A unique ID, as in OSPF.
R1(config-router)# network 172.16.1.0 0.0.0.255	Run EIGRP on the LAN port — the same wildcard idea as OSPF, but no area.
R1(config-router)# network 10.0.0.0 0.0.0.3 R1(config-router)# network 10.0.0.8 0.0.0.3	Run EIGRP on both links.
R1(config-router)# passive-interface g0/0/0	No EIGRP hellos to the LAN.
R1(config-router)# no auto-summary	Advertise the real subnets, not a classful summary.



Check it: **show ip eigrp neighbors** lists R2 and R3; the routes appear as **D ... [90/3072]**.



Watch out: A different AS number (100 vs 10) means no neighbours at all. EIGRP routes show as D, not E.





How to troubleshoot routing

Goal: Work bottom-up: interfaces first, then neighbours, then routes — in both directions.

Symptom	Check with	Likely fix
Interface down	<code>show ip interface brief</code>	no shutdown , the right cable, the same subnet at both ends
No OSPF or EIGRP neighbours	<code>show ip protocols</code> , <code>show ip ospf neighbor</code>	Same area / same AS; correct network statements; router links not passive
A network is missing from the table	<code>show ip route</code> on each router	Add the missing network statement on the router that owns it
Ping works one way only	<code>show ip route</code> on the far router	Add the return route — both directions are needed
Traffic takes an odd path	<code>show ip route</code> (look at the codes)	Lowest distance wins: remove the old method (no router rip)
Ping fails part-way	<code>tracert</code> from a PC	The fault is just after the last router that answered



Tip: Ping each neighbour first: if two directly connected routers can't ping, no routing protocol will work.



08

How to secure switches and routers

Allow only known devices on switch ports, filter traffic with ACLs, and share one public address with NAT. (Lab 6)

In this part, learn how to:

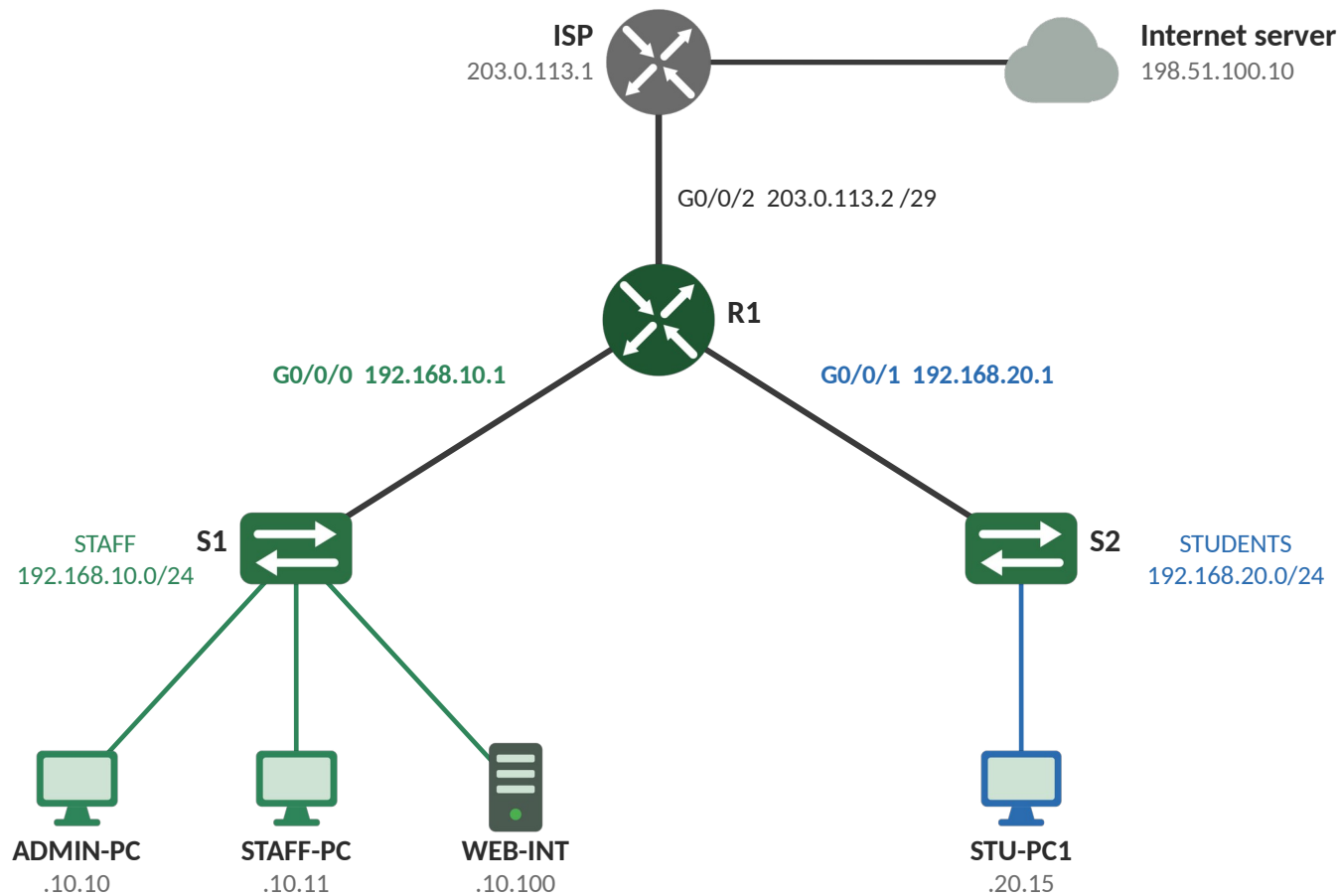
- set up the security lab
- lock switch ports
- understand ACL rules
- write standard and extended ACLs
- test and read an ACL
- configure PAT and static NAT





How to set up the security lab (Lab 6 topology)

Goal: One router joins the STAFF LAN, the STUDENT LAN and the ISP. A written policy says who may reach what.



The policy

1. Only ADMIN-PC may SSH to R1.
2. Students may browse the internal web server — nothing else in STAFF.
3. Everyone reaches the Internet through one public address (PAT).
4. WEB-INT is published to the Internet as 203.0.113.5 (static NAT).
5. Only registered devices may use the staff switch ports.

R1 also needs a default route: `ip route 0.0.0.0 0.0.0.0 203.0.113.1.`





How to lock switch ports with port security

Goal: Only the registered PC may use each staff port. Any other device shuts the port down.

● ● ● S1 — type the white text, then press Enter	What it does
S1(config)# interface range fa0/1 - 3	The three staff ports.
S1(config-if-range)# switchport mode access	Port security only works on a port fixed as access (or trunk), not left dynamic.
S1(config-if-range)# switchport port-security	Turn the feature on (default: 1 MAC address, violation shutdown).
S1(config-if-range)# switchport port-security maximum 1	One device per port. (Use 2 for an IP phone with a PC behind it.)
S1(config-if-range)# switchport port-security mac-address sticky	Learn the first device's MAC and write it into the running-config automatically.
S1(config-if-range)# switchport port-security violation shutdown	An unknown device shuts the port (err-disabled). Other modes: restrict (drop + log), protect (drop quietly).



Check it: Ping from each PC first, then **show port-security interface fa0/2** → Secure-up, 1 sticky MAC.



Watch out: Recover an err-disabled port: remove the rogue device, then **shutdown** and **no shutdown** on the port.

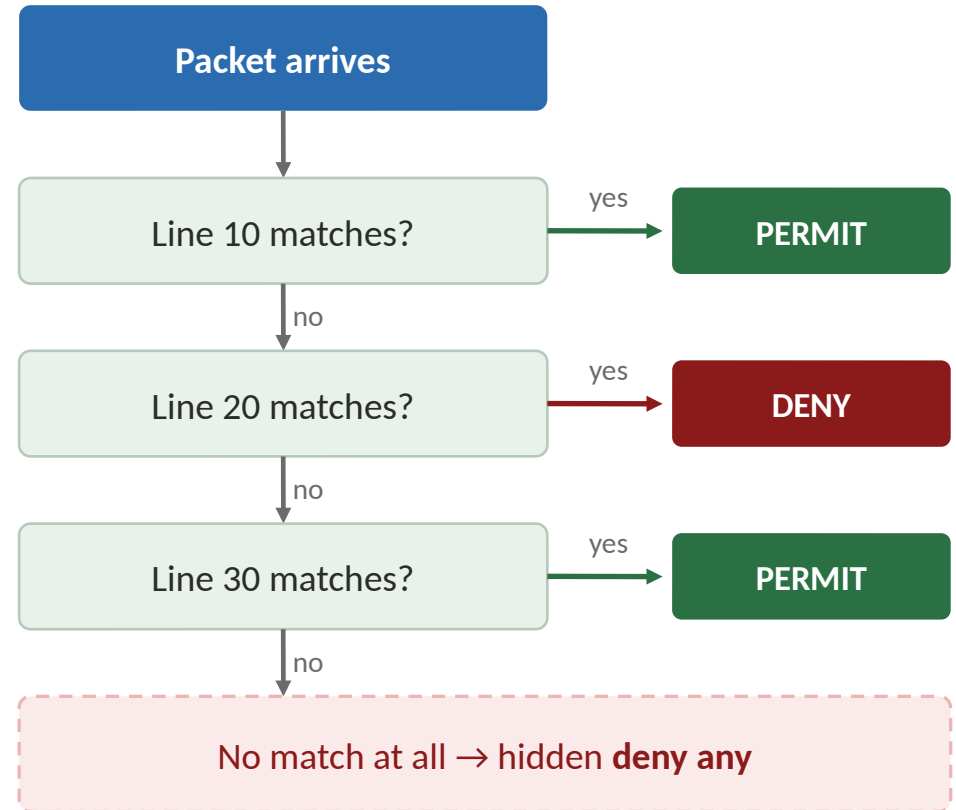




How to understand ACL rules before you write one

Goal: An access control list (ACL) is a list of permit/deny rules that a router checks for every packet.

- Rules are checked **top-down**; the **first match wins** and the rest are skipped.
- Every ACL ends with a hidden **deny any** — include at least one permit.
- **Standard** (1–99): checks the **source** only → place it **near the destination**.
- **Extended** (100–199 or a name): source, destination, protocol and port → place it **near the source**.
- **Wildcards:** 0 = must match, 1 = ignore. **0.0.0.255** = a /24, **host 10.1.1.5** = one address, **any** = all.
- One ACL per interface, per direction (**in** or **out**).



Once a line matches, the packet is permitted or dropped — later lines are never read.





How to let only the admin PC manage the router

Goal: A standard ACL on the VTY lines: only ADMIN-PC (192.168.10.10) may open an SSH session to R1.

R1 — type the white text, then press Enter	What it does
R1(config)# access-list 10 remark ONLY ADMIN-PC MAY MANAGE R1	A note stored with the ACL, for the next administrator.
R1(config)# access-list 10 permit host 192.168.10.10	Permit only the admin PC's address. The hidden deny any blocks everyone else.
R1(config)# line vty 0 4	The remote-access lines.
R1(config-line)# access-class 10 in	Check each incoming SSH session against ACL 10. On lines it is access-class ; on interfaces it is ip access-group .



Check it: ADMIN-PC: **ssh -l netadmin 192.168.10.1** works. STAFF-PC: the connection is refused.



Watch out: Test from ADMIN-PC before you log out — a wrong address locks everyone out (the console still works).





How to filter traffic with an extended named ACL

Goal: Students may open the intranet web server, but nothing else in the STAFF LAN. Everything else is allowed.

R1 — type the white text, then press Enter	What it does
<pre>R1(config)# ip access-list extended STUDENT-POLICY</pre>	Create a named extended ACL. The prompt becomes (config-ext-nacl)# .
<pre>R1(config-ext-nacl)# remark Students may browse the intranet only</pre>	A note stored with the ACL.
<pre>R1(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq 80</pre>	Students (192.168.20.x) may open web pages (TCP port 80) on the web server.
<pre>R1(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255</pre>	Anything else from students to the staff LAN is blocked.
<pre>R1(config-ext-nacl)# permit ip any any</pre>	Everything else (e.g. the Internet) is allowed — this beats the hidden deny any.
<pre>R1(config-ext-nacl)# exit R1(config)# interface g0/0/1 R1(config-if)# ip access-group STUDENT-POLICY in</pre>	Check packets as they enter R1 from the student LAN — close to the source.



Check it: Run the tests on the next slide; **show access-lists** counts the matches of each line.



Watch out: Order matters: if the deny line came first, even the web page would be blocked.





How to test and read an ACL

Goal: Prove the policy with planned tests, then read the match counters to see which line caught each packet.

```
R1 > CLI
R1# show access-lists
Standard IP access list 10
 10 permit host 192.168.10.10 (2 match(es))
Extended IP access list STUDENT-POLICY
 10 permit tcp 192.168.20.0 0.0.0.255 host 192.168.10.100 eq www (4 match(es))
 20 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 (8 match(es))
 30 permit ip any any (12 match(es))
```

Test from STU-PC1	Expected	Because of
Browser: http://192.168.10.100	Page loads	line 10 (TCP 80)
ping 192.168.10.100	Fails	line 20
ping 192.168.10.11	Fails	line 20
Browser: http://198.51.100.10	Page loads	line 30 (after NAT)

Reading the output
eq www = port 80. 10, 20, 30 are sequence numbers: insert a line with **15 permit ...**, remove one with **no 20**.

Think
STAFF-PC → STU-PC1 ping fails too. Why? The reply hits line 20.



How to share one public address with PAT

Goal: All inside PCs reach the Internet through R1's single public address, told apart by port numbers (NAT overload).

● ● ● R1 — type the white text, then press Enter	What it does
<pre>R1(config)# interface g0/0/0 R1(config-if)# ip nat inside</pre>	The STAFF LAN port is on the inside (private addresses).
<pre>R1(config-if)# interface g0/0/1 R1(config-if)# ip nat inside</pre>	The STUDENT LAN port is inside too.
<pre>R1(config-if)# interface g0/0/2 R1(config-if)# ip nat outside R1(config-if)# exit</pre>	The ISP link is the outside (public address 203.0.113.2).
<pre>R1(config)# access-list 1 permit 192.168.10.0 0.0.0.255 R1(config)# access-list 1 permit 192.168.20.0 0.0.0.255</pre>	Choose which inside addresses may be translated. Here the ACL selects — it does not filter.
<pre>R1(config)# ip nat inside source list 1 interface g0/0/2 overload</pre>	Translate those sources to G0/0/2's public address and share it by port number: PAT.



Check it: Browse <http://198.51.100.10> from a PC, then run **show ip nat translations** on R1.



Watch out: NAT “does nothing”? Usually inside and outside are swapped, or the default route to the ISP is missing.





How to publish an internal server with static NAT

Goal: A permanent one-to-one mapping lets Internet users reach the internal web server at a public address.

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# ip nat inside source static 192.168.10.100 203.0.113.5	Internet users who browse to 203.0.113.5 reach WEB-INT (192.168.10.100), and its replies leave as 203.0.113.5.
R1# show ip nat translations	Show the NAT table (below).

```
● ● ● R1 > CLI

Pro  Inside global      Inside local      Outside local      Outside global
---  203.0.113.5          192.168.10.100    ---
tcp  203.0.113.2:1025    192.168.20.15:1025  198.51.100.10:80  198.51.100.10:80
```

Inside local
The private address of the inside host, e.g. 192.168.10.100.

Inside global
The public address it becomes on the Internet, e.g. 203.0.113.5.

Outside local / global
The Internet host — the same in this lab: 198.51.100.10.

✓ **Check it:** From the Internet server’s side, browse <http://203.0.113.5>: the WEB-INT page appears.

⚠ **Watch out:** Use a public address no other device uses — here one from R1’s ISP subnet (203.0.113.0/29).

09

How to add redundancy without loops

Extra links between switches keep users online when a cable fails — if Spanning Tree controls them. EtherChannel turns parallel links into one. (Lab 7)

In this part, learn how to:

- understand Spanning Tree and the lab
- choose the root bridge, PortFast and BPDU Guard
- read show spanning-tree
- bundle links with LACP EtherChannel





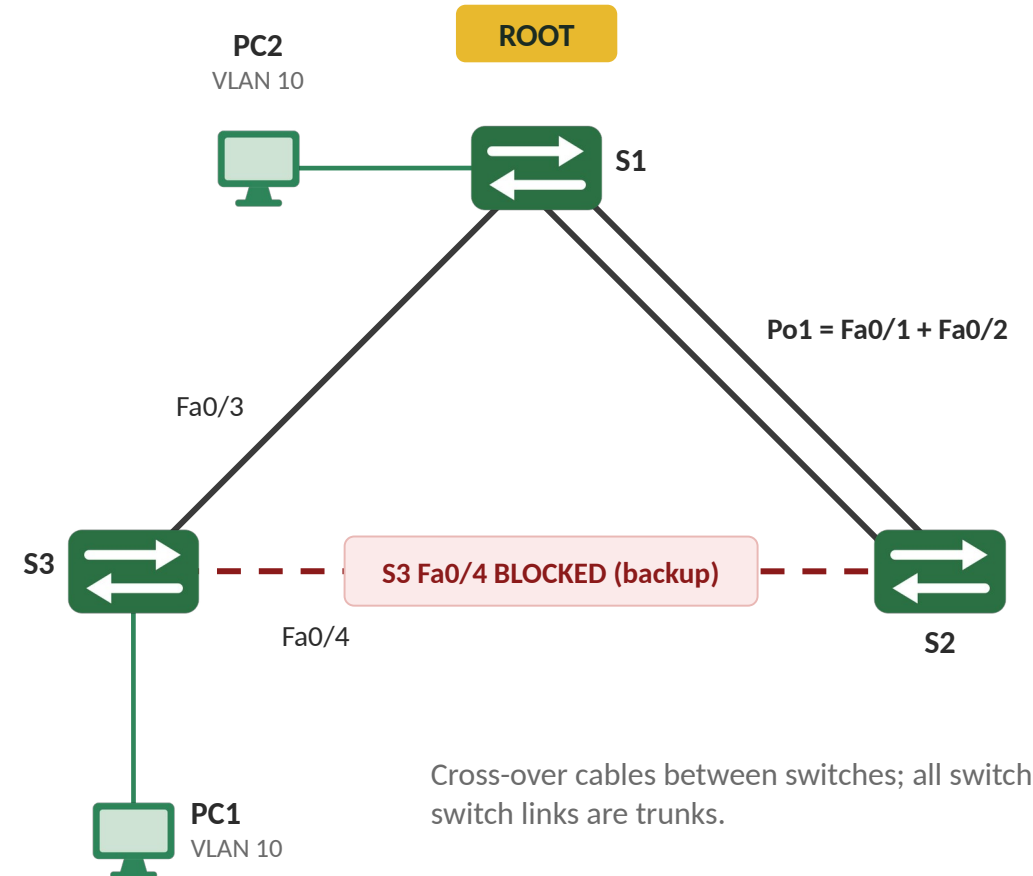
How to understand Spanning Tree before you configure it

Goal: Redundant links are good for uptime but create loops. Spanning Tree keeps one path active and the others on standby.

- **Extra links = backup paths** — but they also create loops.
- **A loop is deadly at Layer 2:** frames have no TTL, so broadcasts circle for ever (a broadcast storm) and the network freezes.
- **STP elects a root bridge** (the lowest bridge ID) and keeps the best path from every switch to it.
- Every other path is **blocked** — but kept ready.
- **If the active path fails**, the blocked port starts forwarding: in 1-2 seconds with Rapid PVST+.

Think of it like this

A ring road with one barrier: traffic never circles, and if the main road closes the barrier lifts.





How to set the root bridge, PortFast and BPDU Guard

Goal: Choose the centre of the network yourself, start PC ports instantly, and protect them from rogue switches.

● ● ● S1 / S2 / S3 — type the white text, then press Enter	What it does
<code>S1(config)# spanning-tree mode rapid-pvst</code>	Use Rapid PVST+: it recovers in about 1–2 s instead of 30–50 s. Type it on every switch.
<code>S1(config)# spanning-tree vlan 1,10,20 root primary</code>	Make S1 the root bridge for these VLANs (priority 24576) — your choice, not luck.
<code>S2(config)# spanning-tree vlan 1,10,20 root secondary</code>	S2 takes over as root if S1 fails (priority 28672).
<code>S3(config)# interface fa0/10</code> <code>S3(config-if)# spanning-tree portfast</code>	The PC port starts forwarding at once — no 30-second wait. Only on ports to end devices!
<code>S3(config-if)# spanning-tree bpduguard enable</code>	If someone plugs a switch into this PC port, the port shuts down (err-disabled) before a loop forms.



Check it: `show spanning-tree vlan 10` on S1 says “This bridge is the root”.



Watch out: Never use PortFast on a port that connects to another switch — it can create a temporary loop.





How to read show spanning-tree

Goal: Find the root, the blocked port and the edge port from one output — then prove that failover works.

S3 › CLI

S3# show spanning-tree vlan 10

VLAN0010

Spanning tree enabled protocol rstp

Root ID Priority **24586**
 Address 0001.4295.A7C1
 Cost 19
 Port 3(FastEthernet0/3)
 Bridge ID Priority 32778

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/3	Root FWD	19	128.3	P2p	
Fa0/4	Altn BLK	19	128.4	P2p	
Fa0/10	Desg FWD	19	128.10	P2p	Edge

Root ID priority 24586

24576 (root primary) + 10 (the VLAN number): the root is S1.

Root FWD

This switch's best path to the root, forwarding. Cost 19 = one 100 Mbps link.

Altn BLK

The blocked backup port. It breaks the loop and waits to take over.

Desg FWD ... Edge

The PC port, forwarding at once thanks to PortFast.

Test: ping -t PC1 → PC2, then shut S3 Fa0/3.





How to bundle links with an LACP EtherChannel

Goal: Join the two S1-S2 links into one logical link: double the bandwidth, and STP blocks neither of them.

S1 — type the white text, then press Enter	What it does
S1(config)# interface range fa0/1 - 2	The two parallel links between S1 and S2.
S1(config-if-range)# channel-group 1 mode active	Bundle them as Port-channel 1 using LACP. active proposes the bundle; the other end must be active or passive .
S1(config-if-range)# exit	Back to global mode. Interface Port-channel1 now exists.
S1(config)# interface port-channel 1 S1(config-if)# switchport mode trunk	From now on configure the bundle, not its member ports. Here it becomes one logical trunk.
! Repeat the same commands on S2	A bundle needs both ends configured.
S1# show etherchannel summary	Look for Po1(SU) (S = Layer 2, U = in use) with both members marked (P) = bundled.



Check it: Both switches show **Po1(SU)** with **Fa0/1(P)** **Fa0/2(P)**; STP now treats Po1 as one link.



Watch out: Members must match: speed, duplex, access/trunk mode and allowed VLANs. (I) or (s) = a mismatch.



10

How to manage devices and add IPv6

Keep every device on the same clock, send logs to a server, discover neighbours, back up configurations — and switch on IPv6. (Labs 9 and 10)

In this part, learn how to:

- set the time and send logs (NTP, Syslog)
- discover neighbours and allow monitoring (CDP, LLDP, SNMP)
- back up and restore with TFTP
- configure basic IPv6 addressing and routing





How to set the time and send logs to a server

Goal: Logs are only useful if every device has the right time and sends its messages to one place.

R1 — type the white text, then press Enter	What it does
<pre>R1# clock set 09:00:00 6 October 2026</pre>	Set the clock by hand (privileged mode). Fine for one device; NTP keeps every device in step.
<pre>R1(config)# ntp server 192.168.50.10</pre>	Take the time from the NTP server (SRV1), so the logs of all devices line up.
<pre>R1(config)# service timestamps log datetime msec</pre>	Stamp every log message with the date and time.
<pre>R1(config)# logging host 192.168.50.10</pre>	Copy every log message to the Syslog service on SRV1.
<pre>R1(config)# logging trap debugging</pre>	Send all severity levels (0–7) — fine in a lab. Real networks usually send informational (6).



Check it: `show ntp status` → “Clock is synchronized”. SRV1 › Services › SYSLOG lists R1’s messages.



Watch out: Switch the services on first: SRV1 › Services › NTP and SYSLOG must both be On.





How to discover neighbours and allow monitoring

Goal: CDP and LLDP show who is connected where; SNMP lets a monitoring tool read each device's status.

R1 — type the white text, then press Enter	What it does
R1# show cdp neighbors	Lists the Cisco devices plugged into R1: name, model, and the local and remote ports that join them.
R1# show cdp neighbors detail	Adds each neighbour's IP address and IOS version — ideal for mapping a network you don't know.
R1(config)# lldp run R1# show lldp neighbors	LLDP is the open-standard version of CDP (works with other vendors). It is off by default.
R1(config)# snmp-server community KNP-RO ro	Monitoring tools that know the community "KNP-RO" may read R1's information. ro = read-only.
R1(config)# snmp-server location KNP-ICT-MDF-Rack1 R1(config)# snmp-server contact ict@knp.ac.ke	Where the device is and who looks after it — shown on monitoring dashboards.



Check it: PC › Desktop › MIB Browser: Advanced... (R1's address, read community KNP-RO), then Get a value.



Watch out: CDP and LLDP reveal device details: switch them off on ports facing the Internet (**no cdp enable**).





How to back up and restore a configuration with TFTP

Goal: Keep a copy of every configuration on a server, so a replaced or broken router is restored in minutes.

```
R1 > CLI (sample)

R1# copy running-config tftp:
Address or name of remote host []? 192.168.50.10
Destination filename [R1-config]? R1-2026-10-06.cfg
Writing running-config...!!
[OK - 1052 bytes]

R1# copy tftp: running-config
Address or name of remote host []? 192.168.50.10
Source filename []? R1-2026-10-06.cfg
Destination filename [running-config]?
! press Enter to accept
```

- 1 **Prepare the server**
SRV1 > Services > TFTP → On.
- 2 **Back up**
copy running-config tftp: → server address → a file name with device and date.
- 3 **Check**
The file appears in SRV1's TFTP file list.
- 4 **Restore**
copy tftp: running-config when a router is replaced or a change goes wrong.



Watch out: TFTP has no password and no encryption — real networks use SCP or SFTP. Back up after every change.





How to configure basic IPv6 on a router (1 of 2)

Goal: Give R1's ports IPv6 addresses and turn on IPv6 routing, so PCs can configure themselves (SLAAC).

● ● ● R1 — type the white text, then press Enter	What it does
R1(config)# ipv6 unicast-routing	Turn on IPv6 routing (off by default). Without it R1 won't forward IPv6 or advertise its prefixes, so SLAAC fails.
R1(config)# interface g0/0/0	The LAN port.
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64	A global unicast address. /64 is the standard prefix for every LAN — SLAAC needs 64 host bits.
R1(config-if)# ipv6 address fe80::1 link-local	A short, easy link-local address. PCs use it as their default gateway.
R1(config-if)# no shutdown	Switch the port on (if it is not on already).
R1(config-if)# interface g0/0/1 R1(config-if)# ipv6 address 2001:db8:acad:12::1/64 R1(config-if)# ipv6 address fe80::1 link-local R1(config-if)# no shutdown	The same on the link to R2. (R2 uses ::2 and fe80::2; its LAN is 2001:db8:acad:2::/64.)



Check it: **show ipv6 interface brief** → each port up/up with its FE80::1 and 2001:DB8:... addresses.



Watch out: Type the colons carefully: :: may appear only once in an address.





How to configure basic IPv6 on a router (2 of 2)

Goal: Add routes between the IPv6 LANs, let the PCs configure themselves, and test end to end.

R1 / R2 / PC1 — type the white text, then press Enter	What it does
<pre>R1(config)# ipv6 route 2001:db8:acad:2::/64 2001:db8:acad:12::2</pre>	A static IPv6 route to R2's LAN via R2's address on the shared link — like IPv4, with a prefix length.
<pre>R2(config)# ipv6 route ::/0 2001:db8:acad:12::1</pre>	An IPv6 default route on R2 (::/0 = every address), back to R1.
<pre>! PC1: IP Configuration > IPv6 > Automatic</pre>	The PC builds its own address from R1's /64 prefix (SLAAC) and uses fe80::1 as its gateway.
<pre>R1# show ipv6 route</pre>	C, L and S routes for IPv6 — read it just like the IPv4 table.
<pre>C:\> ping 2001:db8:acad:2::10</pre>	From PC1, ping PC2 (static 2001:db8:acad:2::10/64, gateway fe80::2).

Check it: `tracert 2001:db8:acad:2::10` from PC1 passes R1 and then R2.

Watch out: SLAAC fails without **ipv6 unicast-routing** — R1 never sends the Router Advertisements PCs need.



11

How to test and troubleshoot your network

A calm, systematic method finds any fault: climb the ping ladder, ask the right show command, fix one thing, re-test, and log it.

In this part, learn how to:

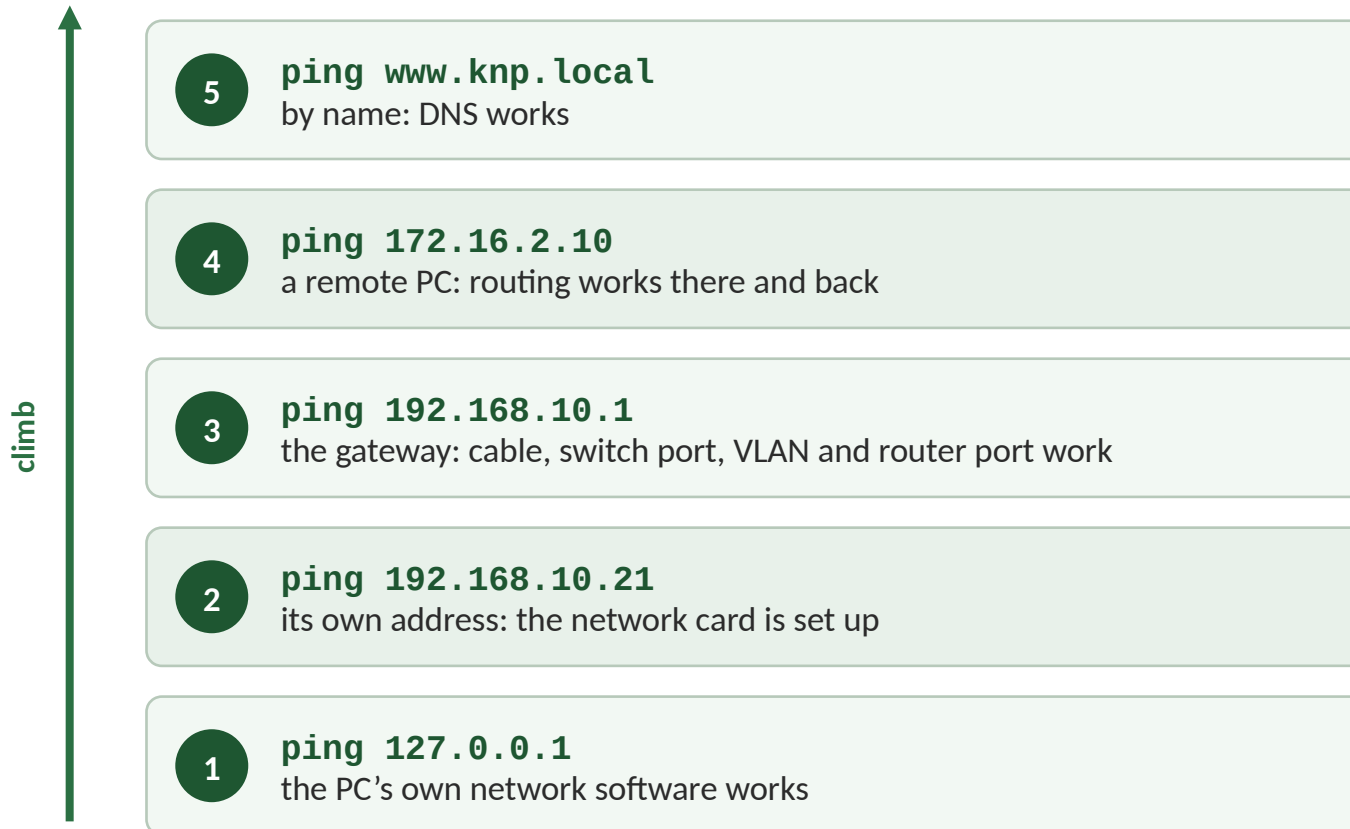
- climb the ping ladder
- choose the right show command
- fix the most common Packet Tracer mistakes
- keep a troubleshooting log





How to troubleshoot step by step: the ping ladder

Goal: Climb from the PC itself to the far network. The first step that fails tells you where the fault is.



Stop at the first step that fails

The fault is between that step and the one below it.

Then follow the method

1. Symptom: what exactly fails?
2. Theory: what could cause it?
3. Test it with a show command
4. Fix one thing only
5. Re-test, then log it

tracert shows the last router that answered — the fault is just after it.





How to choose the right show command

Goal: Each question has one command that answers it. Learn this table and you can diagnose almost anything.

Question	Command	Look for
Are my ports up and addressed?	show ip interface brief	up / up and the right IP
What is configured?	show running-config	typos, a missing no shutdown
Which port is in which VLAN?	show vlan brief	ports beside the right VLAN
Are the trunks working?	show interfaces trunk	trunking, native VLAN, allowed VLANs
Which device is plugged in where?	show cdp neighbors	device, local port, remote port
Where did the switch learn a MAC?	show mac address-table	the MAC on the expected port
Does the router know the way?	show ip route	a C, S, O, R or D route
Did DHCP lease addresses?	show ip dhcp binding	one line per PC
Are OSPF neighbours up?	show ip ospf neighbor	FULL state
Is the ACL matching?	show access-lists	match counts going up
Is NAT translating?	show ip nat translations	inside local ↔ inside global
Is port security blocking?	show port-security interface fa0/1	Secure-up or Secure-shutdown





How to fix the most common Packet Tracer mistakes (1 of 2)

Goal: These faults cause most “it doesn’t work” moments in class. Check them first.

Symptom	Likely cause	Fix
Red link light on a router port	The port was never switched on	no shutdown on that interface
Red link light, port is on	Wrong cable type	Straight-through between different devices; cross-over between the same kind
Amber light on a switch port	Spanning Tree is still starting the port	Wait about 30 s, or click Fast Forward Time
% Invalid input detected	A typo, or the wrong mode	Read the prompt; use ? to see what is allowed
PC cannot ping its own gateway	Wrong IP/mask/gateway, or port in the wrong VLAN	ipconfig on the PC; show vlan brief on the switch
PCs in different VLANs cannot ping	Trunk, sub-interface or gateway problem	show interfaces trunk ; check encapsulation dot1Q and each PC’s gateway



Tip: Fix one thing at a time and re-test — otherwise you won’t know which change fixed it.





How to fix the most common Packet Tracer mistakes (2 of 2)

Goal: More faults to rule out — from DHCP and routing to saving and hardware.

Symptom	Likely cause	Fix
PC address is 169.254.x.x	No DHCP reply	See “How to fix common DHCP problems” (part 06)
Remote network unreachable	A route is missing — often the return route	show ip route on every router along the path
Configuration gone after a reload	It was never saved	copy running-config startup-config after every working step
SSH connection refused	No RSA keys or domain name, or login instead of login local	show ip ssh ; check the VTY lines
G0/0/2 missing on a 4331	The SFP slot is empty	Power off, insert a GLC-T module, power on
The first ping loses one reply	ARP is finding the MAC address	Normal — ping again



Tip: Still stuck? Compare **show running-config** line by line with the slide for that skill.





How to keep a troubleshooting log

Goal: Write down every fault as you go. The log is assessment evidence and stops you fixing the same thing twice.

#	Test / symptom	Command used	Root cause	Fix applied	Re-test
1	PC3 cannot ping PC6	<code>show interfaces trunk</code>	VLAN 30 not allowed on the trunk	<code>switchport trunk allowed vlan add 30</code>	PASS
2					
3					
4					
5					



Watch out: Change one thing at a time, and re-test after each change.



Tip: Lab 8 scoring: 2 points for each fault fixed and logged; minus 1 for any new fault you create.



12

How to practise and revise on your own

Two challenge labs that combine everything, a checklist to judge your own competence, and one-page command cards for revision.

In this part, learn how to:

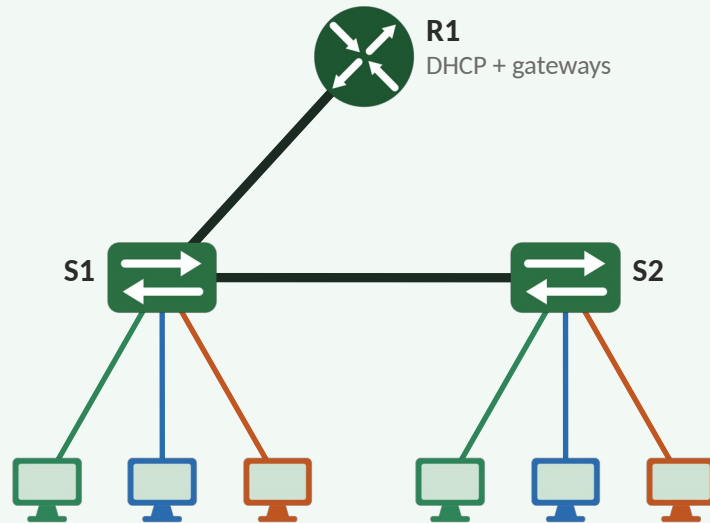
- build challenge lab A: VLANs, routing between them and DHCP
- build challenge lab B: routing, NAT and security
- check yourself against the competence list
- revise with the switch and router cards





How to practise: challenge lab A — VLANs, routing and DHCP

Goal: Rebuild Lab 2 without the slides. Time yourself: aim for under 90 minutes.



VLAN 10 / 20 / 30 PCs on both switches; VLAN 99 = management

Tasks — from a blank workspace

1. Basic set-up and SSH on R1, S1 and S2; shut unused ports.
2. VLANs 10 STAFF, 20 STUDENTS, 30 FINANCE, 99 MGMT on both switches.
3. Access ports as in the plan; trunks with native VLAN 99.
4. Router-on-a-stick on R1 G0/0/1 (.10, .20, .30, .99 native).
5. DHCP pools for VLANs 10, 20, 30 (exclude .1-.20); PCs on DHCP.
6. Switch management in VLAN 99; save every device.

You are competent when...

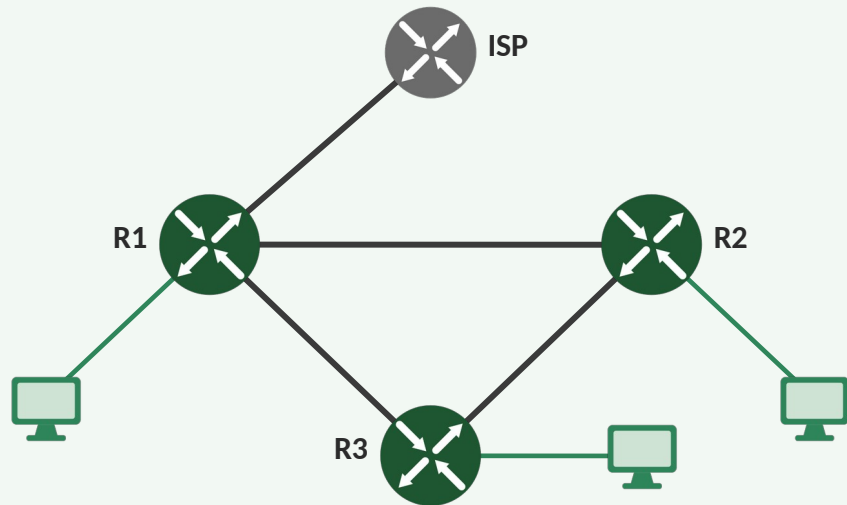
every PC leases an address from its own VLAN, PCs in different VLANs ping each other, **show interfaces trunk** shows native VLAN 99, SSH works and Telnet fails.





How to practise: challenge lab B — routing, NAT and security

Goal: Combine Lab 3 and Lab 6: OSPF inside, PAT to the ISP, and a security policy.



Lab 3 triangle + an ISP router on R1 (203.0.113.0/29)

Tasks — from a blank workspace

1. Address all links (/30) and LANs; OSPF area 0 on R1, R2, R3.
2. R1: default route to the ISP + **default-information originate**.
3. PAT on R1 for all three LANs (inside: LAN and links; outside: ISP).
4. ACL: LAN 3 may reach LAN 1 only by web (TCP 80).
5. SSH on every router; VTY access only from one admin PC.
6. Port security on the LAN 1 switch; NTP and Syslog to a server.

You are competent when...

every PC reaches the ISP side, **show ip nat translations** shows the PCs, the ACL tests pass, OSPF neighbours are FULL, and the failover of one link is survived.





How to check you are competent: a self-assessment list

Goal: Tick each line only when you can do it without the slides — this is what the practical assessment looks for.

- | | |
|--|---|
| ☐ Choose and connect the right cables | ☐ Route between VLANs on a Layer 3 switch |
| ☐ Move between IOS modes and use ? | ☐ Configure DHCP pools, exclusions and relay |
| ☐ Do the basic set-up and save it | ☐ Configure static, default and OSPF routes |
| ☐ Address router ports and a switch SVI | ☐ Lock ports with port security |
| ☐ Set up SSH with local users | ☐ Write and apply standard and extended ACLs |
| ☐ Create VLANs and access ports | ☐ Configure PAT and static NAT |
| ☐ Configure trunks with a native VLAN | ☐ Set the root bridge, PortFast, EtherChannel |
| ☐ Configure router-on-a-stick | ☐ Find faults with the ping ladder and show commands |





How to remember the commands: switch quick reference

Goal: One page of the switch commands used in this workbook. The prompt shows the mode; replace the example values.

Basic set-up

```
hostname S1
enable secret <password>
line console 0
  password <password>
  login
interface vlan 99
  ip address <ip> <mask>
ip default-gateway <router-ip>
copy running-config startup-config
```

VLANs and trunks

```
vlan 10
  name STAFF
interface range fa0/1 - 10
  switchport mode access
  switchport access vlan 10
interface g0/1
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan add 40
```

Security, STP, EtherChannel

```
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
spanning-tree mode rapid-pvst
spanning-tree vlan 10 root primary
spanning-tree portfast
spanning-tree bpduguard enable
channel-group 1 mode active
```

Check it

```
show vlan brief
show interfaces trunk
show ip interface brief
show mac address-table
show port-security interface fa0/1
show spanning-tree vlan 10
show etherchannel summary
show cdp neighbors
```





How to remember the commands: router quick reference

Goal: One page of the router commands used in this workbook. Practise until you no longer need it.

Interfaces and SSH

```
interface g0/0/0
  ip address <ip> <mask>
  no shutdown
interface g0/0/1.10
  encapsulation dot1Q 10
  ip domain-name <domain>
crypto key generate rsa general-keys modulus 1024
line vty 0 4
  transport input ssh
  login local
```

Routing

```
ip route <network> <mask> <next-hop>
ip route 0.0.0.0 0.0.0.0 <isp-address>
router ospf 10
  router-id 1.1.1.1
  network <network> <wildcard> area 0
  passive-interface g0/0/0
router eigrp 100
router rip
ipv6 unicast-routing
```

DHCP, NAT, ACL

```
ip dhcp excluded-address <first> <last>
ip dhcp pool <NAME>
  network <network> <mask>
  default-router <gateway>
ip helper-address <server>
ip nat inside | ip nat outside
ip nat inside source list 1 interface g0/0/2 overload
ip access-group <ACL> in
```

Check it

```
show ip interface brief
show ip route
show ip protocols
show ip ospf neighbor
show ip dhcp binding
show ip nat translations
show access-lists
show running-config
```





How to keep improving after this workbook

Competence comes from repetition: rebuild each lab from a blank workspace until you can do it without the slides.

Questions?



Repeat

Rebuild every lab from scratch; time yourself.



Save evidence

Keep each .pkt file and config backup in your portfolio.



Read further

Cisco Networking Academy: Packet Tracer and CCNA courses; Jeremy's IT Lab (free CCNA videos).



Ask

Bring every error message to class — they are the best lessons.

